Predicting unplanned readmission for decompensation within one year in hepatitis B related decompensated cirrhosis

Online Resource 1

Hepatology International

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Supplementary methods, extended results, Tables S1–S9, and Figs. S1–S2. References use the numbering in the main article.

Note M1 Variable definitions coding and measurement

Tables S1 and S2 describe the Chinese and external cohorts. Table S3A lists 50 recorded characteristics and identifies the 46 predefined candidate predictors, with their definitions, units, analytical roles, and availability. Hepatic encephalopathy, prothrombin time, prothrombin activity, and MELD-Na were descriptive or benchmark variables. Obesity (BMI ≥28 kg/m²), hypertension, and diabetes mellitus were descriptive components of the metabolic-comorbidity composite and were not separate feature-selection candidates.

Baseline descriptions of the Chinese cohort use pre-imputation records and variable-specific nonmissing denominators. For HBV DNA, 205 results were quantitatively measured and 232 were left-censored with numerical bounds of 20, 50, 100, or 1,000 IU/mL. Half the recorded bound was substituted before $\log_{10}$ transformation. Five further results recorded as not detected without a numerical bound had no continuous value but were assigned the lowest category; three had no recorded result. The continuous and categorical denominators were therefore 437 and 442. Left-censoring was distinguished from ordinary missingness.

HBsAg was analyzed on the $\log_{10}$ IU/mL scale. The historical analytical convention used 0.05 and replaced recorded zero values with 0.025. This convention does not identify a verified universal assay detection limit or imply uniform treatment of all results below 0.05.

Multiple imputation in the Chinese cohort was performed after the development–validation split using development-estimated parameters. Five completed datasets were generated, and one fixed completed dataset was used for modeling. The retained historical imputation record identifies the second completed dataset and predictive mean matching for neutrophil, lymphocyte and monocyte counts and continuous HBV DNA. Validation data did not estimate imputation parameters; outcomes were neither imputed nor included as predictors of missing values. Derived indices used the formulas and rounding rules in Table S3A. Supplementary descriptive summaries use the original pre-imputation data.

External evaluation comprised 102 eligible hospitalization episodes from 50 patients, including 67 event and 35 non-event episodes. Inputs were observed measurements from the same continuous emergency–inpatient encounter, selected through an admission-oriented hierarchy without statistical imputation. Of 816 component selections, 804 fell within −24 to +24 hours of admission, nine after 24 and within 48 hours, and three followed the predefined extended hierarchy beyond 48 hours. Results had to be available by the applicable deadline and index discharge. Thus, measurements were closely aligned to admission but were not universally restricted to 48 hours. Table S2 specifies episode-level, patient-level, and variable-specific denominators. SMDs describe case mix. Clinical review and database-based postdischarge ascertainment are detailed in Note M6.

Note M2 Fixed models nested evaluation and patient aggregation

LASSO at λ_{1se} and Boruta with TentativeRoughFix independently evaluated all 46 predefined candidates in the development cohort. Their intersection defined Core4: INR, PLT, ALBI, and FIB-4. Parallel univariable logistic analyses did not filter either selection input. Spearman correlations and VIFs were assessed after selection (Table S3B and C). Table S4 gives final model settings and search configurations; final fixed hyperparameters do not specify those selected in every nested partition.

The repeated nested evaluation comprised 100 outcome-stratified partitions, each with 312 outer training and 133 outer test patients. Within each partition, chained-equation imputation generated five datasets over 20 iterations; the second completed dataset was used for selection and fitting. The imputation method and predictor matrix were initialized from outer training data, and outer test records were excluded from parameter estimation using the ignore setting. Outcomes were neither imputed nor used as imputation predictors. Missing HBV DNA, neutrophil, lymphocyte, and monocyte values used predictive mean matching; derived indices were recalculated afterward. LASSO and Boruta selection, tuning, and fitting then used outer training data only. All 100 selection procedures completed. The primary seven-model comparison used 98 compatible partitions; LASSO and elastic net had 95 valid evaluations because three additional compatible partitions yielded a single predictor unsupported by their fitting interface. Failed or excluded fits were not counted as valid results. Table S5A gives medians and quartiles across the actual valid partitions, which describe resampling variability rather than CIs.

For nested LASSO selection, continuous candidates were min–max scaled using outer training values; a binomial glmnet model used tenfold cross-validation, 1,000 candidate penalty values, standardize = false, and λ_{1se}. Boruta used pValue = 0.01, multiplicity adjustment, maxRuns = 1,000, and random-forest Z-score importance, followed by TentativeRoughFix where needed. These settings describe the nested selection procedure; Table S4 separately records the fixed development-model settings.

For the nested RBF-SVM–MELD-Na comparison, 98 paired outer test partitions contributed predictions. Each patient's risk was the mean of predictions obtained only when that patient was held out. All 445 Chinese patients (185 events) were represented, with 15–45 held-out predictions each (median, 29). The MELD-Na logistic benchmark was fitted within the corresponding outer training partitions. Table S5D presents this patient-level aggregate separately from partition-specific distributions. Paired patient-bootstrap intervals condition on the aggregated predictions; model development was not repeated within bootstrap samples. The aggregate remains an internal evaluation of the Chinese cohort.

Note M3 Predictor representations ablations and event timing

Table S6A compares three representations using logistic regression and RBF-SVM in both fixed evaluation cohorts. Original Core4 comprised INR, PLT, ALBI, and FIB-4. The raw-variable candidate pool retained BMI and excluded length of stay; its selected intersection was INR and PLT. The composite-variable pool retained 11 derived indices and excluded their ten raw components, leaving 36 candidates; its intersection was PTAR and FIB-4. Length of stay remained a candidate in the composite analysis but was not selected. The composite pool therefore included a variable available only after admission. AUROC, AP, Brier score, CITL, and calibration slope are reported as point estimates.

Table S6B evaluates six ablations: E1 omitted INR; E2 omitted PLT; E3 omitted ALBI; E4 omitted FIB-4; E5 retained INR and ALBI; and E6 retained PLT and FIB-4. ΔAUROC and ΔAP are the full model minus the ablated model, whereas Brier improvement is the ablated model minus the full model. Positive differences favor the full model. Reduced models were fitted in the original 312-patient development cohort without repeating feature selection; full-model predictions were unchanged.

CIs were obtained from paired differences, rather than subtraction of marginal intervals, and condition on the evaluated predictions without repeating model development. These exploratory comparisons were unadjusted for multiplicity. Intervals crossing zero do not establish equivalence or absence of a predictor contribution; ablations evaluate prediction rather than independent causal effects.

Table S7 evaluates scores from the six original fixed 1-year models, without fitting separate early or late models. Early events occurred at 0–90 days, and later events after 90 and within 365 days. Each event group was compared with the same cohort-specific 1-year non-event controls; the other event group was excluded from that comparison. Internal counts were 22 early events, 33 later events, and 78 shared controls. External counts were 48 early events, 17 later events, and 35 shared controls. Two external event episodes with unclassifiable timing remained in the annual analysis but were excluded from both timing comparisons.

The external early comparison included 83 episodes from 48 patients; the later comparison included 52 episodes from 37 patients. These overlapping sets cannot be summed to reconstruct the primary cohort. The 5,000 full-cohort bootstrap samples resampled patients internally and patient clusters externally, with early and later contrasts evaluated in the same resamples. CIs and later-minus-earlier differences retain their paired definitions. These analyses assess event-time-stratified discrimination of annual scores; they are not cumulative 90-day ROC analyses or estimates of time-specific calibrated probabilities.

Note M4 Clinical benchmarking calibration and decision measures

The fixed MELD-Na benchmark was a development-fitted single-predictor logistic model applied unchanged to both evaluation cohorts. Table S5B reports absolute performance and diagnostic calibration; Table S5C reports fixed paired comparisons. Calibration fits were diagnostic and did not replace predictions. Higher AUROC and AP and lower Brier scores indicate better performance. Ideal CITL is zero; a positive CITL indicates overall underprediction. Ideal calibration slope and observed-to-expected ratio (O/E) are one. AP is not positive predictive value at a single threshold.

Fig. 4d and e group observations separately for each model using the original evaluation sample's type 7 prediction quantiles: five groups internally and four externally. Duplicate boundaries are merged, and equal predictions are not split across groups. Points show group mean predicted probabilities and observed event proportions. Group membership remains fixed during resampling. Vertical error bars show 95% percentile bootstrap intervals for group observed proportions, using 5,000 patient resamples internally and patient-cluster resamples externally; they are distinct from the Wilson intervals in Fig. 4c. Horizontal error bars are not plotted.

Calibration curves are diagnostic logistic fits of the observed outcome on the logit of the original predicted probability, with a freely estimated intercept and slope. Curves connect saved coordinates only within each model's observed probability range; curve confidence bands are not displayed. The free intercept used for the curve differs from CITL, which estimates an intercept with the logit prediction as an offset and its coefficient fixed at one. Probabilities are clipped to $[10^{-6}, 1 - 10^{-6}]$ only for logit-based diagnostic calculations. These fits characterize calibration and do not recalibrate or replace the predictions used for performance assessment.

Fig. 4c and Table S9 use tertiles of the 312 development-cohort RBF-SVM predictions, calculated using type 7 quantiles without optimizing outcome separation. The fixed cutoffs are 0.234116093557522 and 0.559109361430369: low risk, $p \leq$ the lower cutoff; intermediate risk, lower cutoff $< p \leq$ upper cutoff; high risk, $p >$ the upper cutoff. Event-rate intervals are Wilson intervals internally and patient-cluster bootstrap intervals externally. Coverage is the fraction of eligible observations selected; event capture is the fraction of all observed events among selected observations; missed events occur outside that group. These cutoffs describe the development risk distribution and do not define validated treatment or follow-up thresholds.

Fig. S1 compares three models at nominal capacities of 10%, 20%, 30%, 40%, and 50%. Each strategy selected $K = \text{floor}(qN)$ observations, where q is nominal capacity and N is the record count; actual coverage K/N can therefore differ from q. Ranking ties followed a fixed outcome-blind rule. Pointwise intervals used 5,000 resamples, repeating selection within each resample. Resampling used patients internally and patient clusters retaining all eligible episodes externally. Intervals condition on the fixed predictions.

Fig. S2 shows decision thresholds from 0.05 to 0.80, including treat-all, treat-none, negative net benefit, and intervals crossing zero. Paired differences are RBF-SVM minus either MELD-Na or Core4 logistic

regression. Models shared the same bootstrap samples within each cohort, with patient clustering externally; intervals were calculated from within-sample contrasts. Intervals are pointwise rather than simultaneous across thresholds. Neither an optimal threshold nor an optimal capacity was selected. Positive estimated net benefit concerns a hypothetical action and requires prospective evaluation before it can imply clinical benefit.

Note M5 Prediction attribution and reference background

Exact Kernel SHAP used kernelshap version 0.9.1 on the positive-class probability scale. The fixed RBF-SVM and its prediction wrapper were retained. For each explained record, all four-feature coalitions were evaluated against the same empirical background of 312 locked development records, without coalition sampling. The common baseline model output was 0.416286863888495. The background matrix differed from the original fitted-model input in one recorded FIB-4 value, which was 7.12 in the reference matrix and 7.13 in the original training matrix; the model was not refitted and neither matrix was altered for the attribution calculation. The background therefore represents the locked development reference rather than a byte-identical copy of the original fit matrix.

Saved attribution checks confirmed agreement with the fixed-model predictions and local additivity. The maximum absolute error in baseline + ΣSHAP − prediction was $1.11022302462516 \times 10^{-16}$ in both evaluation cohorts. An independent explicit-coalition implementation checked three internal records selected solely by proximity to the 10th, 50th, and 90th percentiles of predicted risk; the current external calculations also underwent an independent direct check. These numerical checks are distinct from the two illustrative cases displayed in Fig. 5g and h, which were selected after inspection of predictions and outcomes.

The attributions are exact relative to the specified empirical reference and probability function, rather than exact biological effects. Background-based feature replacement can produce combinations outside observed physiological relationships, particularly because PLT is a component of FIB-4. Attribution patterns should be interpreted jointly and do not establish causality or monotonicity over all possible predictor values.

Table S1 Outcome-stratified characteristics of the Chinese cohort

A Development cohort

Characteristic	No target readmission (n = 182)	Target readmission (n = 130)	P value
Demographic and background characteristics			
Age (years)	51.0 (44.0, 60.0)	55.0 (47.0, 63.0)	0.029
Male sex	132 (72.5)	87 (66.9)	0.286
Body mass index (kg/m²)	22.42 (20.32, 24.45)	22.59 (20.99, 24.43)	0.440
Metabolic comorbidities	55 (30.2)	38 (29.2)	0.851
Smoking history	78 (42.9)	54 (41.5)	0.816
Alcohol use history	77 (42.3)	49 (37.7)	0.413
HBV-related virological characteristics			
HBeAg/anti-HBe status			0.971
HBeAg-/anti-HBe+	127 (69.8)	88 (67.7)	
HBeAg+/anti-HBe-	32 (17.6)	24 (18.5)	
HBeAg-/anti-HBe-	15 (8.2)	11 (8.5)	
HBeAg+/anti-HBe+	8 (4.4)	7 (5.4)	
HBV DNA study category (IU/mL)	n = 182	n = 127	0.001
<2,000	120 (65.9)	59 (46.5)	
2,000-<20,000	17 (9.3)	12 (9.4)	
≥20,000	45 (24.7)	56 (44.1)	
HBV DNA, log10 (IU/mL)	2.70 (2.70, 4.38); n = 178	3.81 (2.70, 5.83); n = 126	0.021
HBsAg, log10 (IU/mL)	2.76 (1.90, 3.09)	2.87 (2.17, 3.16)	0.086
Prior antiviral therapy	105 (57.7)	73 (56.2)	0.787
Recorded clinical manifestations			
Ascites	69 (37.9)	61 (46.9)	0.111
EGVH	109 (59.9)	64 (49.2)	0.062
SBP	37 (20.3)	28 (21.5)	0.795
HE	4 (2.2)	19 (14.6)	<0.001
HRS	1 (0.5)	1 (0.8)	1.000
Infection	66 (36.3)	56 (43.1)	0.224

Characteristic	No target readmission (n = 182)	Target readmission (n = 130)	P value
Laboratory measurements			
Albumin (g/L)	31.00 (27.00, 34.00)	28.00 (25.00, 32.00)	<0.001
Total bilirubin (μmol/L)	20.05 (13.93, 31.18)	34.15 (22.15, 50.30)	<0.001
ALT (U/L)	36.00 (26.00, 51.75)	45.00 (31.00, 71.50)	0.004
AST (U/L)	40.00 (29.00, 55.75)	49.00 (33.25, 98.00)	0.001
ALP (U/L)	91.00 (68.00, 118.50)	96.00 (68.00, 143.50)	0.110
GGT (U/L)	34.00 (23.25, 64.50)	40.00 (23.00, 80.50)	0.253
INR	1.28 (1.18, 1.39)	1.48 (1.37, 1.63)	<0.001
PT (s)	15.60 (14.50, 17.00)	17.80 (16.12, 19.38)	<0.001
PTA (%)	66.00 (58.02, 75.80)	53.30 (46.00, 59.00)	<0.001
Platelet count (10^9/L)	77.00 (54.00, 132.00)	51.00 (37.00, 72.50)	<0.001
WBC (10^9/L)	4.25 (2.85, 6.89)	3.43 (2.48, 5.26)	0.003
Hemoglobin (g/L)	89.00 (70.25, 108.00)	89.50 (72.00, 111.00)	0.496
Serum creatinine (μmol/L)	68.00 (56.25, 79.00)	64.00 (55.00, 77.75)	0.273
Sodium (mmol/L)	139.00 (137.00, 141.00)	139.00 (137.00, 142.00)	0.927
Composite scores			
MELD-Na	10.87 (9.20, 13.49)	14.19 (11.83, 17.20)	<0.001
ALBI	-1.81 (-2.10, -1.39)	-1.38 (-1.77, -1.04)	<0.001
FIB-4	4.38 (2.70, 7.01)	9.60 (5.06, 13.91)	<0.001
Index hospitalization			
Length of index hospital stay (days)	10.0 (7.0, 14.0)	11.0 (7.0, 15.0)	0.451

B Fixed internal validation cohort

Characteristic	No target readmission (n = 78)	Target readmission (n = 55)	P value
Demographic and background characteristics			
Age (years)	53.5 (47.0, 59.0)	50.0 (45.0, 60.0)	0.261
Male sex	55 (70.5)	39 (70.9)	0.961
Body mass index (kg/m²)	21.86 (20.31, 23.22)	23.39 (21.50, 25.52)	0.003
Metabolic comorbidities	23 (29.5)	15 (27.3)	0.781
Smoking history	35 (44.9)	18 (32.7)	0.159
Alcohol use history	35 (44.9)	22 (40.0)	0.576

Characteristic	No target readmission (n = 78)	Target readmission (n = 55)	P value
HBV-related virological characteristics			
HBeAg/anti-HBe status			0.461
HBeAg-/anti-HBe+	56 (71.8)	42 (76.4)	
HBeAg+/anti-HBe-	11 (14.1)	10 (18.2)	
HBeAg-/anti-HBe-	8 (10.3)	2 (3.6)	
HBeAg+/anti-HBe+	3 (3.8)	1 (1.8)	
HBV DNA study category (IU/mL)			0.623
<2,000	48 (61.5)	33 (60.0)	
2,000-<20,000	12 (15.4)	6 (10.9)	
≥20,000	18 (23.1)	16 (29.1)	
HBV DNA, log10 (IU/mL)	2.70 (2.70, 4.16)	2.70 (2.70, 4.58)	0.357
HBsAg, log10 (IU/mL)	2.71 (1.88, 3.14)	2.91 (2.01, 3.28)	0.139
Prior antiviral therapy	39 (50.0)	29 (52.7)	0.757
Recorded clinical manifestations			
Ascites	40 (51.3)	24 (43.6)	0.385
EGVH	42 (53.8)	31 (56.4)	0.774
SBP	23 (29.5)	20 (36.4)	0.404
HE	1 (1.3)	3 (5.5)	0.306
HRS	3 (3.8)	0 (0.0)	0.267
Infection	33 (42.3)	26 (47.3)	0.570
Laboratory measurements			
Albumin (g/L)	31.00 (27.00, 36.00)	27.00 (25.00, 32.50)	0.002
Total bilirubin (μmol/L)	21.60 (13.75, 35.42)	32.80 (18.05, 49.20)	0.005
ALT (U/L)	38.50 (27.25, 51.75)	41.00 (28.00, 56.50)	0.819
AST (U/L)	42.50 (31.25, 62.25)	45.00 (30.00, 74.00)	0.766
ALP (U/L)	92.50 (67.00, 129.00)	90.00 (65.50, 137.00)	0.980
GGT (U/L)	55.00 (27.25, 108.75)	40.00 (23.00, 65.50)	0.069
INR	1.26 (1.19, 1.41)	1.52 (1.35, 1.71)	<0.001
PT (s)	15.75 (15.03, 17.20)	18.30 (16.00, 20.05)	<0.001
PTA (%)	66.40 (57.15, 75.00)	53.00 (44.50, 62.30)	<0.001

Characteristic	No target readmission (n = 78)	Target readmission (n = 55)	P value
Platelet count (10^9/L)	67.00 (54.25, 94.75)	51.00 (36.00, 66.50)	0.001
WBC (10^9/L)	4.07 (2.78, 5.64)	4.06 (2.87, 5.54)	0.866
Hemoglobin (g/L)	97.50 (78.25, 115.75)	95.00 (79.00, 108.00)	0.264
Serum creatinine (μmol/L)	71.00 (60.25, 83.75)	66.00 (58.00, 74.50)	0.038
Sodium (mmol/L)	138.00 (136.00, 140.75)	138.00 (136.00, 141.00)	0.927
Composite scores			
MELD-Na	11.71 (9.29, 14.74)	14.33 (12.25, 17.57)	<0.001
ALBI	-1.73 (-2.17, -1.39)	-1.36 (-1.74, -1.03)	<0.001
FIB-4	5.04 (2.88, 7.99)	7.26 (5.07, 11.15)	0.005
Index hospitalization			
Length of index hospital stay (days)	11.5 (8.0, 19.0)	10.0 (8.0, 14.0)	0.371

Values are medians (Q1, Q3) or n (%) from available pre-imputation records, with percentages based on nonmissing denominators. Available HBV DNA n in the development non-event/event groups was 178/126 for the continuous measure and 182/127 for the categorical measure. All displayed analytical fields were available in both fixed internal-validation groups.

P values compare outcome groups within each cohort using two-sided Mann–Whitney U tests for continuous variables and chi-squared or Fisher exact tests for categorical variables. Comparisons were exploratory and unadjusted for multiplicity and did not determine entry into feature selection. Clinical manifestations can coexist.

Metabolic comorbidity denotes obesity, hypertension, or diabetes mellitus; obesity was defined as BMI ≥28 kg/m². HBeAg/anti-HBe status represents the four combinations of positive and negative results. Viral-marker coding and measurement limitations are detailed in Table S3A.

Abbreviations: HBV, hepatitis B virus; DNA, deoxyribonucleic acid; HBeAg, hepatitis B e antigen; anti-HBe, antibody to hepatitis B e antigen; HBsAg, hepatitis B surface antigen; EGVH, esophagogastric variceal hemorrhage; SBP, spontaneous bacterial peritonitis; HE, hepatic encephalopathy; HRS, hepatorenal syndrome; ALT, alanine aminotransferase; AST, aspartate aminotransferase; ALP, alkaline phosphatase; GGT, gamma-glutamyl transferase; INR, international normalized ratio; PT, prothrombin time; PTA, prothrombin activity; WBC, white blood cell count; MELD-Na, Model for End-Stage Liver Disease-sodium; ALBI, albumin-bilirubin score; FIB-4, fibrosis-4 index; Q1/Q3, first/third quartiles.

Table S2 External cohort characteristics and case-mix comparison

A External cohort structure and index-hospitalization profile

Characteristic	External cohort
Cohort structure	
Eligible index hospitalization episodes	102
Unique patients	50
Eligible episodes per patient	1.00 [1.00, 2.75]
Patients contributing 1 eligible episode	34 (68.0%)
Patients contributing 2 eligible episodes	3 (6.0%)
Patients contributing ≥3 eligible episodes	13 (26.0%)
1-year readmission events	67/102 (65.7%)
Demographics	
Age, years	58.00 [54.00, 65.00]
Male sex	86/102 (84.3%)
Race/ethnicity, n (%)	
Asian	26 (25.5%)
Black	12 (11.8%)
Hispanic/Latino	10 (9.8%)
White	52 (51.0%)
Other/unknown	2 (2.0%)
Index hospitalization characteristics	
Hospital length of stay, days	4.16 [2.55, 6.89]
Any ICU admission during index hospitalization	13/102 (12.7%)
ICU length of stay among ICU-exposed episodes, days	1.80 [0.89, 2.10]; n = 13
Index decompensation manifestations, n/N (%)	
Ascites	84/101 (83.2%)
Hepatic encephalopathy	26/101 (25.7%)
Variceal bleeding	7/101 (6.9%)
Spontaneous bacterial peritonitis	8/101 (7.9%)
Hepatorenal syndrome	2/101 (2.0%)

B Chinese and external case-mix comparison

Characteristic	Chinese cohort (N = 445 patients)	External (N = 102 episodes)	SMD
Demographics			
Age, years	52.00 [45.00, 61.00]	58.00 [54.00, 65.00]	-0.582
Male sex	313/445 (70.3%)	86/102 (84.3%)	-0.339
Liver and laboratory profile			
Albumin, g/L	29.00 [26.00, 34.00]	29.00 [26.00, 34.00]	0.036
Total bilirubin, μmol/L	24.70 [15.70, 41.00]	35.91 [17.10, 63.27]	-0.411
INR	1.37 [1.22, 1.52]	1.50 [1.40, 1.70]	-0.504
Platelet count, ×10^9/L	63.00 [44.00, 95.00]	67.50 [42.50, 101.75]	-0.076
Creatinine used for MELD-Na, μmol/L	67.00 [56.00, 79.00]	88.40 [70.72, 112.71]	-0.606
Sodium used for MELD-Na, mmol/L	139.00 [137.00, 141.00]	135.00 [131.00, 138.75]	0.921
Composite liver-disease indices			
ALBI score	-1.62 [-2.02, -1.22]	-1.49 [-1.95, -1.09]	-0.165
FIB-4	5.76 [3.38, 10.53]	9.65 [5.60, 15.39]	-0.484
MELD-Na	12.47 [9.99, 15.54]	18.43 [13.89, 23.69]	-1.044

Values are medians (Q1, Q3) or n/N (%) unless otherwise stated. Hospitalization episodes are the analysis unit, and patients are the clustering unit for uncertainty estimation. Patient-contribution percentages use 50 patients; race/ethnicity percentages use 102 episodes. ICU length of stay is summarized across the 13 ICU-exposed episodes.

Index decompensation manifestations were identified through the MIMIC-specific structured-data adjudication. Manifestations can coexist; denominators reflect available phenotype-support information. Race/ethnicity categories were collapsed from recorded MIMIC-IV categories for description.

SMD is Chinese minus external using pooled standard deviations; positive values indicate higher Chinese-cohort values. SMDs describe case mix and were not used for hypothesis testing or model selection. Chinese-cohort measures reflected index-admission status; external measures were aligned to admission as closely as availability allowed. Case-mix comparisons do not isolate causes of performance differences.

The admission-oriented measurement hierarchy and its exceptions are described in Note M1.

Abbreviations: ICU, intensive care unit; MIMIC, Medical Information Mart for Intensive Care; SMD, standardized mean difference; INR, international normalized ratio; ALBI, albumin-bilirubin score; FIB-4, fibrosis-4 index; MELD-Na, Model for End-Stage Liver Disease-sodium; Q1/Q3, first/third quartiles. Core4 comprises INR, platelet count, ALBI and FIB-4.

Table S3 Predictor definitions availability and selection

A Predictor definitions and availability profiles

Variable	Unit/scale	Role	Definition/coding	Profile
Metabolic comorbidities	Category	Candidate	Obesity OR hypertension OR diabetes mellitus	A
EGVH	Category	Candidate	Stored definition	A
Ascites	Category	Candidate	Stored definition	A
SBP	Category	Candidate	Stored definition	A
HE	Category	Description only	Stored definition	A
HRS	Category	Candidate	Stored definition	A
Infection	Category	Candidate	Stored definition	A
HBeAg/anti-HBe status	Category	Candidate	Stored definition	A
Sex	Category	Candidate	Stored definition	A
Prior antiviral therapy	Category	Candidate	Stored definition	A
Smoking history	Category	Candidate	Stored definition	A
Alcohol use history	Category	Candidate	Stored definition	A
Serum creatinine (μmol/L)	μmol/L	Candidate	Stored definition	A
Total bilirubin (μmol/L)	μmol/L	Candidate	Stored definition	A
ALT (U/L)	U/L	Candidate	Stored definition	A
AST (U/L)	U/L	Candidate	Stored definition	A
GGT (U/L)	U/L	Candidate	Stored definition	A
LDH (U/L)	U/L	Candidate	Stored definition	A
ALP (U/L)	U/L	Candidate	Stored definition	A
Albumin (g/L)	g/L	Candidate	Stored definition	A
PTA (%)	%	Description only	Stored definition	A
PT (s)	s	Description only	Stored definition	A
Hemoglobin (g/L)	g/L	Candidate	Stored definition	A
NEU (10^9/L)	10^9/L	Candidate	Stored definition	B
LYM (10^9/L)	10^9/L	Candidate	Stored definition	B
MON (10^9/L)	10^9/L	Candidate	Stored definition	B
Platelet count (10^9/L)	10^9/L	Candidate; Core4	Stored definition	A
WBC (10^9/L)	10^9/L	Candidate	Stored definition	A

Variable	Unit/scale	Role	Definition/coding	Profile
Potassium (mmol/L)	mmol/L	Candidate	Stored definition	A
Sodium (mmol/L)	mmol/L	Candidate	Stored definition	A
Chloride (mmol/L)	mmol/L	Candidate	Stored definition	A
INR	ratio	Candidate; Core4	Stored definition	A
HBsAg, log10 (IU/mL)	log10 (IU/mL)	Candidate	Viral-marker coding below	A
HBV DNA study category (IU/mL)	IU/mL cutpoints	Candidate	0: <2,000; 1: 2,000-<20,000; 2: ≥20,000 IU/mL	C
HBV DNA, log10 (IU/mL)	log10 (IU/mL)	Candidate	Viral-marker coding below	D
Age (years)	years	Candidate	Stored definition	A
MELD-Na	score	Description only	Formula below	A
ALBI	score	Candidate; Core4	Formula below	A
PMR	Formula-defined index/ratio	Candidate	Formula below	B
PTAR	Formula-defined index/ratio	Candidate	Formula below	A
NAR	Formula-defined index/ratio	Candidate	Formula below	B
NLR	Formula-defined index/ratio	Candidate	Formula below	B
FIB-4	score	Candidate; Core4	Formula below	A
LMR	Formula-defined index/ratio	Candidate	Formula below	B
PLR	Formula-defined index/ratio	Candidate	Formula below	B
SIRI	Formula-defined index/ratio	Candidate	Formula below	B
PNI	Formula-defined index/ratio	Candidate	Formula below	B
APRI	Formula-defined index/ratio	Candidate	Formula below	A
Length of index hospital stay (days)	days	Candidate	Stored definition	A
Body mass index	kg/m²	Candidate	Stored definition	A

All 50 recorded characteristics are listed; 46 were candidate predictors. HE, PT, PTA, and MELD-Na were descriptive or benchmark variables. Core4 comprises INR, PLT, ALBI, and FIB-4. Non-viral continuous variables retain their recorded numerical scales. Clinical categories reflect recorded classifications; measurement times and treatment timing were not assumed to be uniform.

A Pre-imputation availability by cohort

Profile	Overall N = 445	Development N = 312	Internal N = 133
A	445 / 0	312 / 0	133 / 0
B	438 / 7	306 / 6	132 / 1
C	442 / 3	309 / 3	133 / 0
D	437 / 8	304 / 8	133 / 0

Cells show available n / missing n. The profile assigned to each variable specifies its pre-imputation counts. Numerically encoded left-censored results are not counted as ordinary missingness.

A Pre-imputation availability by outcome group

Profile	Development no readmission N = 182	Development readmission N = 130	Internal no readmission N = 78	Internal readmission N = 55
A	182 / 0	130 / 0	78 / 0	55 / 0
B	178 / 4	128 / 2	77 / 1	55 / 0
C	182 / 0	127 / 3	78 / 0	55 / 0
D	178 / 4	126 / 4	78 / 0	55 / 0

Cells show available n / missing n before imputation. Completed modeling datasets constitute a separate data state.

A Metabolic-comorbidity components

Component	Unit/scale	Definition
Obesity	Binary (0/1)	Derived from BMI ≥28 kg/m².
Hypertension	Binary (0/1)	Recorded structured diagnosis flag.
Diabetes mellitus	Binary (0/1)	Recorded structured diagnosis flag.

These variables describe components of the metabolic-comorbidity composite and were not separate feature-selection candidates.

A Derived predictor definitions

Predictor	Formula
MELD-Na	Scr_mg_dl = min(4, max(1, Scr/88.4)); TBIL_mg_dl = max(1, TBIL/17.1); M = 3.78 × ln(TBIL_mg_dl) + 11.2 × ln(max(1, INR)) + 9.57 × ln(Scr_mg_dl) + 6.43; Na = min(137, max(125, Sodium)); round(M + 1.32 × (137-Na) - 0.033 × M × (137-Na), 2).
ALBI	round(0.66 × log10(TBIL) - 0.085 × Albumin, 2)
PMR	PLT/MON (MON > 0); no rounding
PTAR	10 × INR/Albumin (Albumin > 0); no rounding
NAR	10 × NEU/Albumin (Albumin > 0); no rounding
NLR	round(NEU/LYM, 2)
FIB-4	round(Age × AST / (PLT × sqrt(ALT)), 2)
LMR	round(LYM/MON, 2)

Predictor	Formula
PLR	round(PLT/LYM, 2)
SIRI	round(NEU × MON/LYM, 2)
PNI	round(Albumin + 5 × LYM, 2)
APRI	round((AST/40) × 100/PLT, 2)

Albumin is in g/L, bilirubin and creatinine in μmol/L, and cell counts in 10^9/L. The study-specific definition PTAR = 10 × INR/albumin is used.

A Additional descriptions Overall

Variable	Overall (N = 445)
LDH (U/L)	388.00 (288.00, 520.00)
NEU (10^9/L)	2.68 (1.75, 4.74); n=438
LYM (10^9/L)	0.69 (0.47, 1.02); n=438
MON (10^9/L)	0.31 (0.20, 0.46); n=438
Potassium (mmol/L)	3.80 (3.50, 4.20)
Chloride (mmol/L)	107.00 (104.00, 109.00)
PMR	221.72 (143.92, 322.14); n=438
PTAR	0.45 (0.38, 0.57)
NAR	0.93 (0.57, 1.69); n=438
NLR	3.69 (2.50, 6.75); n=438
LMR	2.40 (1.67, 3.42); n=438
PLR	92.52 (63.41, 143.15); n=438
SIRI	1.08 (0.56, 2.64); n=438
PNI	34.02 (29.81, 37.85); n=438
APRI	1.73 (1.01, 3.27)

Values are medians (Q1, Q3), with available n when it differs from the group total. Outcome-group P values are exploratory, unadjusted, and were not used for feature selection.

A Additional descriptions Development

Variable	No target readmission (N = 182)	Target readmission (N = 130)	P value
LDH (U/L)	381.00 (292.00, 519.75)	401.50 (296.00, 545.25)	0.460
NEU (10^9/L)	3.08 (1.84, 5.39); n=178	2.32 (1.50, 3.61); n=128	0.001
LYM (10^9/L)	0.76 (0.50, 1.15); n=178	0.65 (0.47, 0.95); n=128	0.026
MON (10^9/L)	0.32 (0.21, 0.48); n=178	0.27 (0.19, 0.44); n=128	0.071
Potassium (mmol/L)	3.90 (3.52, 4.20)	3.80 (3.50, 4.10)	0.222

Variable	No target readmission (N = 182)	Target readmission (N = 130)	P value
Chloride (mmol/L)	106.00 (104.00, 109.00)	108.00 (104.00, 110.75)	0.015
PMR	254.06 (169.15, 393.45); n=178	180.70 (131.34, 278.33); n=128	<0.001
PTAR	0.42 (0.36, 0.49)	0.52 (0.44, 0.62)	<0.001
NAR	1.01 (0.60, 1.84); n=178	0.83 (0.53, 1.39); n=128	0.040
NLR	3.54 (2.50, 6.67); n=178	3.56 (2.38, 6.18); n=128	0.551
LMR	2.44 (1.77, 3.50); n=178	2.38 (1.73, 3.44); n=128	0.696
PLR	102.17 (65.17, 156.44); n=178	75.50 (54.46, 115.70); n=128	<0.001
SIRI	1.19 (0.57, 2.76); n=178	0.94 (0.49, 2.00); n=128	0.075
PNI	35.35 (31.52, 39.62); n=178	31.88 (28.84, 35.76); n=128	<0.001
APRI	1.39 (0.78, 2.27)	2.69 (1.56, 4.74)	<0.001

Values are medians (Q1, Q3), with available n when it differs from the group total. Outcome-group P values are exploratory, unadjusted, and were not used for feature selection.

A Additional descriptions Internal validation

Variable	No target readmission (N = 78)	Target readmission (N = 55)	P value
LDH (U/L)	391.50 (282.00, 550.50)	384.00 (267.00, 499.00)	0.502
NEU (10^9/L)	2.53 (1.79, 4.72); n=77	3.03 (1.93, 4.58)	0.663
LYM (10^9/L)	0.71 (0.46, 1.00); n=77	0.57 (0.46, 0.93)	0.227
MON (10^9/L)	0.33 (0.22, 0.47); n=77	0.31 (0.17, 0.44)	0.569
Potassium (mmol/L)	3.90 (3.52, 4.30)	3.80 (3.60, 4.20)	0.460
Chloride (mmol/L)	105.00 (103.00, 108.00)	107.00 (105.00, 109.50)	0.045
PMR	233.33 (168.42, 308.33); n=77	178.12 (114.95, 287.14)	0.032
PTAR	0.42 (0.37, 0.48)	0.55 (0.44, 0.69)	<0.001
NAR	0.90 (0.56, 1.39); n=77	1.08 (0.70, 1.85)	0.129
NLR	3.71 (2.52, 7.11); n=77	4.70 (2.83, 7.74)	0.295
LMR	2.33 (1.67, 3.33); n=77	2.25 (1.25, 3.40)	0.600
PLR	107.04 (71.05, 153.13); n=77	78.43 (61.36, 119.92)	0.021
SIRI	1.03 (0.65, 2.36); n=77	1.08 (0.67, 2.94)	0.631
PNI	34.95 (30.90, 39.65); n=77	31.75 (28.30, 35.55)	0.003

Variable	No target readmission (N = 78)	Target readmission (N = 55)	P value
APRI	1.38 (0.86, 2.66)	2.21 (1.42, 3.60)	0.006

Values are medians (Q1, Q3), with available n when it differs from the group total. Outcome-group P values are exploratory, unadjusted, and were not used for feature selection.

A HBV DNA measurement status

Status	Overall n	Analytical handling
Quantified result	205	Recorded continuous value and category
Left-censored with a recorded bound	232	Half of the recorded bound before log10 transformation
Not detected; numerical bound unavailable	5	Continuous value missing; original lowest-category assignment retained
No HBV DNA result	3	Continuous and categorical fields unavailable

The 445 observations describe measurement and encoding states. Recorded left-censoring bounds were 20, 50, 100, and 1,000 IU/mL.

A Viral-marker coding and limitations

Field	Scale/definition	Measurement limitation
HBV DNA	log10 (IU/mL); categories <2,000, 2,000-<20,000 and ≥20,000 IU/mL	Continuous and categorical availability differ. Medians and rank-based tests do not remove uncertainty from censoring or substitution.
HBsAg	Stored log10 (IU/mL); available 445/445	The historical 0.05-IU/mL convention and zero-to-0.025 coding are not verified assay-wide detection limits; no further substitution was applied.

Pre-imputation categorical HBV DNA summaries use recorded categories; completed modeling categories use the completed $\log_{10}$ HBV DNA values.

B Fixed-development selection status

Predictor	LASSO λmin	LASSO λ1se	Original Boruta	After RoughFix	Core4
Metabolic comorbidities	False	False	Rejected	False	False
EGVH	False	False	Rejected	False	False
Ascites	True	False	Rejected	False	False
SBP	True	False	Rejected	False	False
HRS	False	False	Rejected	False	False
Infection	True	False	Rejected	False	False
HBeAg status	True	False	Rejected	False	False
Sex	False	False	Rejected	False	False
Prior antiviral therapy	False	False	Rejected	False	False

Predictor	LASSO λmin	LASSO λ1se	Original Boruta	After RoughFix	Core4
Smoking history	True	False	Rejected	False	False
Alcohol use history	True	False	Rejected	False	False
Scr	False	False	Rejected	False	False
TBIL	False	False	Confirmed	True	False
ALT	False	False	Rejected	False	False
AST	False	False	Rejected	False	False
GGT	False	False	Rejected	False	False
LDH	False	False	Rejected	False	False
ALP	False	False	Rejected	False	False
Albumin	False	False	Tentative	True	False
Hb	False	False	Rejected	False	False
NEU	False	False	Rejected	False	False
LYM	False	False	Rejected	False	False
MON	False	False	Rejected	False	False
PLT	True	True	Confirmed	True	True
WBC	False	False	Rejected	False	False
Potassium	False	False	Rejected	False	False
Sodium	False	False	Rejected	False	False
Chloride	False	False	Confirmed	True	False
INR	True	True	Confirmed	True	True
HBsAg	False	False	Rejected	False	False
HBV DNA (3 groups)	True	True	Rejected	False	False
HBV DNA (numeric)	False	False	Rejected	False	False
Age	True	False	Rejected	False	False
ALBI	True	True	Confirmed	True	True
PMR	False	False	Confirmed	True	False
PTAR	False	False	Confirmed	True	False
NAR	False	False	Rejected	False	False
NLR	False	False	Rejected	False	False
FIB-4	True	True	Confirmed	True	True
LMR	False	False	Rejected	False	False
PLR	False	False	Rejected	False	False

Predictor	LASSO λmin	LASSO λ1se	Original Boruta	After RoughFix	Core4
SIRI	False	False	Rejected	False	False
PNI	False	False	Rejected	False	False
APRI	False	False	Confirmed	True	False
Length of stay	False	False	Rejected	False	False
BMI	False	False	Rejected	False	False

Both methods evaluated all 46 candidate source variables in the 312-patient development cohort, without filtering by univariable P values. Statuses refer to source variables rather than individual design-matrix terms. Initial Boruta results were nine confirmed, one tentative, and 36 rejected variables. Albumin was tentative and was retained after TentativeRoughFix. LASSO retained five variables at λıse, and Boruta retained ten; their intersection defined Core4.

B Full nested-selection frequencies

Predictor	Core4 reference set	Nested LASSO n (%)	Nested Boruta n (%)	Nested intersection n (%)
Metabolic comorbidities	No	6 (6%)	0 (0%)	0 (0%)
EGVH	No	9 (9%)	0 (0%)	0 (0%)
Ascites	No	20 (20%)	0 (0%)	0 (0%)
SBP	No	13 (13%)	0 (0%)	0 (0%)
HRS	No	0 (0%)	0 (0%)	0 (0%)
Infection	No	10 (10%)	0 (0%)	0 (0%)
HBeAg status	No	10 (10%)	0 (0%)	0 (0%)
Sex	No	11 (11%)	0 (0%)	0 (0%)
Prior antiviral therapy	No	4 (4%)	0 (0%)	0 (0%)
Smoking history	No	8 (8%)	0 (0%)	0 (0%)
Alcohol use history	No	26 (26%)	0 (0%)	0 (0%)
Scr	No	0 (0%)	16 (16%)	0 (0%)
TBIL	No	0 (0%)	99 (99%)	0 (0%)
ALT	No	0 (0%)	6 (6%)	0 (0%)
AST	No	0 (0%)	0 (0%)	0 (0%)
GGT	No	0 (0%)	19 (19%)	0 (0%)
LDH	No	1 (1%)	3 (3%)	0 (0%)
ALP	No	0 (0%)	10 (10%)	0 (0%)
Albumin	No	0 (0%)	50 (50%)	0 (0%)
Hb	No	0 (0%)	0 (0%)	0 (0%)
NEU	No	0 (0%)	2 (2%)	0 (0%)
LYM	No	0 (0%)	10 (10%)	0 (0%)

Predictor	Core4 reference set	Nested LASSO n (%)	Nested Boruta n (%)	Nested intersection n (%)
MON	No	0 (0%)	6 (6%)	0 (0%)
PLT	Yes	67 (67%)	100 (100%)	67 (67%)
WBC	No	0 (0%)	0 (0%)	0 (0%)
Potassium	No	0 (0%)	1 (1%)	0 (0%)
Sodium	No	0 (0%)	7 (7%)	0 (0%)
Chloride	No	1 (1%)	56 (56%)	1 (1%)
INR	Yes	100 (100%)	100 (100%)	100 (100%)
HBsAg	No	0 (0%)	3 (3%)	0 (0%)
HBV DNA (categorical)	No	98 (98%)	2 (2%)	2 (2%)
log10 HBV DNA	No	0 (0%)	8 (8%)	0 (0%)
Age	No	0 (0%)	1 (1%)	0 (0%)
ALBI	Yes	85 (85%)	97 (97%)	85 (85%)
PMR	No	0 (0%)	70 (70%)	0 (0%)
PTAR	No	5 (5%)	100 (100%)	5 (5%)
NAR	No	0 (0%)	0 (0%)	0 (0%)
NLR	No	0 (0%)	0 (0%)	0 (0%)
FIB-4	Yes	46 (46%)	100 (100%)	46 (46%)
LMR	No	0 (0%)	0 (0%)	0 (0%)
PLR	No	0 (0%)	63 (63%)	0 (0%)
SIRI	No	0 (0%)	2 (2%)	0 (0%)
PNI	No	0 (0%)	42 (42%)	0 (0%)
APRI	No	1 (1%)	100 (100%)	1 (1%)
Length of stay	No	1 (1%)	0 (0%)	0 (0%)
BMI	No	0 (0%)	0 (0%)	0 (0%)

All 100 outer training selections completed, and selection frequencies use all 100 regardless of subsequent model-fitting failures. Each outcome-stratified 70:30 partition comprised 312 outer training and 133 outer test patients. Selection used LASSO at λ_{1se} and Boruta with TentativeRoughFix. Original Core4 membership served as a reference and did not restrict nested selection.

C Post-selection Spearman correlations (development n = 312)

Predictor	INR	PLT	ALBI	FIB-4
INR	1.000	-0.351	0.518	0.434
PLT	-0.351	1.000	-0.143	-0.744
ALBI	0.518	-0.143	1.000	0.343
FIB-4	0.434	-0.744	0.343	1.000

C Post-selection linear-redundancy diagnostics (development n = 312)

Predictor	VIF	Tolerance
INR	1.490	0.671
Platelet count	1.295	0.772
ALBI	1.407	0.711
FIB-4	1.343	0.744

VIF = $1/(1 - R^2)$, obtained by regressing each Core4 predictor on the other three; tolerance = 1/VIF. These post-selection diagnostics were not selection criteria. They describe correlation and residual linear redundancy, rather than biological independence.

C Joint Core4 retention and similarity

Summary	Value	Definition
All four Core4 predictors retained	35/100 (35%)	All four retained, allowing other predictors
Exact Core4 recovery	33/100 (33%)	Selected intersection exactly Core4
Jaccard similarity to original Core4	0.75 (0.50, 1.00)	Median (Q1, Q3); intersection size divided by union size

Frequencies and quartiles describe behavior across repeated overlapping partitions; they are not independent-sample CIs or estimates of causal importance.

Abbreviations: EGVH, esophagogastric variceal hemorrhage; SBP, spontaneous bacterial peritonitis; HE, hepatic encephalopathy; HRS, hepatorenal syndrome; INR, international normalized ratio; PLT, platelet count; TBIL, total bilirubin; ALT, alanine aminotransferase; AST, aspartate aminotransferase; ALP, alkaline phosphatase; GGT, gamma-glutamyl transferase; LDH, lactate dehydrogenase; PT, prothrombin time; PTA, prothrombin activity; WBC, white blood cell count; Hb, hemoglobin; Scr, serum creatinine; NEU, neutrophil count; LYM, lymphocyte count; MON, monocyte count; ALBI, albumin-bilirubin score; FIB-4, fibrosis-4 index; MELD-Na, Model for End-Stage Liver Disease-sodium; PMR, platelet-to-monocyte ratio; PTAR, study-defined 10 × INR/albumin; NAR, neutrophil-to-albumin ratio, calculated as 10 × NEU/albumin; NLR, neutrophil-to-lymphocyte ratio; LMR, lymphocyte-to-monocyte ratio; PLR, platelet-to-lymphocyte ratio; SIRI, systemic inflammation response index; PNI, prognostic nutritional index; APRI, aspartate aminotransferase-to-platelet ratio index; LASSO, least absolute shrinkage and selection operator; VIF, variance inflation factor. HBV, hepatitis B virus; DNA, deoxyribonucleic acid; HBeAg, hepatitis B e antigen; anti-HBe, antibody to hepatitis B e antigen; HBsAg, hepatitis B surface antigen; Q1/Q3, first/third quartiles.

Table S4 Configuration of the seven candidate models

A Fixed final model parameters and preprocessing

Model	Fixed final parameters	Preprocessing	Development search
Logistic regression	link = logit; use_pred_offset = true	Not explicit in learner parameter set	No parameter search
LASSO	alpha = 1; lambda = 0.0232294425776311; standardize = true; use_pred_offset = true	Standardize = true (glmnet)	10-fold CV × 5 repeats; 30 MBO evaluations; optimize AUROC
Elastic net	alpha = 0.0842256563195503; lambda = 0.0893884745834839; standardize = true; use_pred_offset = true	Standardize = true (glmnet)	10-fold CV × 5 repeats; 30 MBO evaluations; optimize AUROC
Random forest	importance = none; min.node.size = 29; mtry.ratio = 0.2; num.threads = 1; num.trees = 1000; sample.fraction = 0.652165659546858	Not explicit in learner parameter set	10-fold CV × 5 repeats; 30 MBO evaluations; optimize AUROC
XGBoost	alpha = 0; colsample_bytree = 1; eta = 0.0184324067769678; gamma = 0; lambda = 0.655478463773341; max_depth = 1; min_child_weight = 4.42216015579396; nrounds = 200; nthread = 1; subsample = 0.661144082310602; verbose = 0; verbosity = 0; use_pred_offset = true	Not explicit in learner parameter set	10-fold CV × 5 repeats; 30 MBO evaluations; optimize AUROC
RBF-SVM	cost = 0.274080705074942; gamma = 0.0573174496538081; kernel = radial; scale = true; type = C-classification	Scale = true	10-fold CV × 5 repeats; 30 MBO evaluations; optimize AUROC
LightGBM	num_leaves = 3; max_depth = 2; learning_rate = 0.014700166690139731; min_data_in_leaf = 50; feature_fraction = 0.75304694004165273; bagging_fraction = 0.70025569698636059; lambda_l2 = 19.999999999999996; nrounds = 169	Identity on raw-scale Core4 values	10-fold CV × 5 repeats (50 inner resamples); 30 MBO evaluations; optimize AUROC

All fixed models used the 312-patient development cohort, with predictors ordered ALBI, FIB-4, INR, and PLT in the fitted learners. The positive class in the six original models was encoded as Yes. Tuning used development data alone. Glmnet standardization and SVM scaling used training-derived transformations; other preprocessing is reported only where specified in the learner settings. For RBF-SVM and Core4 logistic regression, probability prediction used the frozen wrapper with inputs matched by name; the SVM's stored probability-mapping parameters were retained.

Final parameter values describe the fixed development fits. Nested evaluation repeated selection, tuning, and fitting within each outer training partition, rather than imposing the fixed hyperparameters on every partition.

B LightGBM search ranges

Parameter	Search range	Fixed selected value
num_leaves	2 to 15 (integer)	3
max_depth	2 to 5 (integer)	2
learning_rate	0.01 to 0.15 (log scale)	0.014700166690139731
min_data_in_leaf	10 to 50 (integer)	50
feature_fraction	0.7 to 1.0	0.75304694004165273
bagging_fraction	0.7 to 0.95	0.70025569698636059
lambda_l2	0.5 to 20 (log scale)	19.999999999999996
nrounds	50 to 400 (integer)	169

C LightGBM implementation and random settings

Setting	Configuration
Implementation	LightGBM 4.7.0; binary objective; gradient-boosted decision trees; AUROC tuning metric
Inner evaluation	10-fold cross-validation repeated 5 times (50 inner resamples); 30 MBO parameter evaluations
Execution	CPU; num_threads = 1; deterministic = true; force_col_wise = true; force_row_wise = false
Other fixed settings	bagging_freq = 1; feature_pre_filter = false; lambda_l1 = 0; min_gain_to_split = 0; extra_trees = false
Fixed tuning and final-fit seeds	Tuning: 700123; final fit: 1400123
Native random seeds	seed: 800123; data: 900123; feature fraction: 1000123; bagging: 1100123; drop: 1200123; extra: 1300123

Statistical analyses primarily used R 4.5.1 and Python 3.12.14. CV, cross-validation; MBO, model-based optimization. AUROC, area under the receiver operating characteristic curve; LASSO, least absolute shrinkage and selection operator; RBF-SVM, radial-basis-function support vector machine; XGBoost, Extreme Gradient Boosting; LightGBM, Light Gradient Boosting Machine; INR, international normalized ratio; PLT, platelet count; ALBI, albumin-bilirubin score; FIB-4, fibrosis-4 index. Core4 comprises INR, PLT, ALBI and FIB-4. SVM denotes RBF-SVM.

Table S5 Supplementary model performance and clinical benchmarking

A. Repeated nested internal performance

Model	Valid n	AUROC	AP	Brier
Logistic regression	98	0.776 (0.757, 0.797)	0.693 (0.666, 0.723)	0.192 (0.183, 0.201)
LASSO	95	0.776 (0.759, 0.799)	0.692 (0.668, 0.722)	0.192 (0.184, 0.200)
Elastic net	95	0.776 (0.756, 0.799)	0.692 (0.662, 0.723)	0.195 (0.188, 0.205)
Random forest	98	0.764 (0.737, 0.781)	0.645 (0.617, 0.682)	0.196 (0.188, 0.209)
XGBoost	98	0.766 (0.742, 0.780)	0.661 (0.633, 0.692)	0.195 (0.189, 0.205)
RBF-SVM	98	0.777 (0.756, 0.798)	0.688 (0.656, 0.719)	0.193 (0.184, 0.201)
LightGBM	98	0.762 (0.738, 0.779)	0.653 (0.626, 0.688)	0.199 (0.191, 0.209)

Values are medians (Q1, Q3) across valid overlapping partitions and summarize resampling variability rather than CIs.

A. Calibration in the same nested evaluations

Model	Valid n	CITL	Calibration slope
Logistic regression	98	-0.009 (-0.114, 0.064)	0.881 (0.718, 1.098)
LASSO	95	-0.007 (-0.108, 0.054)	0.993 (0.802, 1.323)
Elastic net	95	-0.012 (-0.077, 0.046)	1.196 (0.914, 1.869)
Random forest	98	-0.015 (-0.100, 0.077)	0.861 (0.697, 1.023)
XGBoost	98	-0.025 (-0.087, 0.059)	1.039 (0.850, 1.209)
RBF-SVM	98	-0.101 (-0.196, 0.005)	0.932 (0.775, 1.107)
LightGBM	98	-0.020 (-0.076, 0.043)	1.159 (0.879, 1.595)

A. Execution denominators

Model	Planned	Eligible for primary comparison	Successful (primary set)	Failed (primary set)
Logistic regression	100	98	98	0
LASSO	100	98	95	3
Elastic net	100	98	95	3
Random forest	100	98	98	0
XGBoost	100	98	98	0
RBF-SVM	100	98	98	0
LightGBM	100	98	98	0

Selection completed in all 100 planned partitions. Two partitions with unsupported categorical-predictor selections were excluded from the comparable set for all models. Three additional single-predictor partitions failed for LASSO and elastic net because their fitting interface required at least two columns.

Across all 100 planned partitions, successful/failed fits were 100/0 for logistic regression, 95/5 for LASSO, 95/5 for elastic net, 100/0 for random forest, 98/2 for XGBoost, 98/2 for RBF-SVM, and 100/0 for LightGBM. Completed fits excluded from the primary comparison are distinct from failed fits. Failures were not imputed, replaced, or retried.

B. Fixed model performance and calibration: internal validation

Metric	RBF-SVM	Core4 logistic	MELD-Na benchmark
AUROC	0.751	0.762	0.691
AP	0.672	0.716	0.620
Brier	0.199	0.193	0.218
CITL (95% CI)	0.020 (-0.419 to 0.430)	-0.063 (-0.493 to 0.355)	-0.103 (-0.493 to 0.267)
Calibration slope (95% CI)	0.822 (0.484 to 1.252)	0.796 (0.480 to 1.268)	0.863 (0.442 to 1.413)
O/E (95% CI)	1.009 (0.827 to 1.199)	0.973 (0.801 to 1.148)	0.950 (0.770 to 1.133)
Mean predicted probability (95% CI)	0.410 (0.369 to 0.452)	0.425 (0.381 to 0.468)	0.435 (0.405 to 0.466)
Observed event proportion (95% CI)	0.414 (0.331 to 0.496)	0.414 (0.331 to 0.496)	0.414 (0.331 to 0.496)

AUROC, AP, and Brier score are point estimates without marginal CIs in this panel. Calibration and prediction-summary intervals are reported where available.

B. Fixed model performance and calibration: external evaluation

Metric	RBF-SVM	Core4 logistic	MELD-Na benchmark
AUROC	0.770	0.712	0.669
AP	0.853	0.785	0.732
Brier	0.199	0.225	0.214
CITL (95% CI)	0.725 (0.159 to 1.272)	0.781 (0.085 to 1.458)	0.055 (-0.730 to 0.682)
Calibration slope (95% CI)	0.853 (0.415 to 1.491)	0.223 (-0.038 to 0.774)	0.412 (0.022 to 0.931)
O/E (95% CI)	1.251 (1.058 to 1.447)	1.231 (1.027 to 1.467)	1.015 (0.811 to 1.178)
Mean predicted probability (95% CI)	0.525 (0.432 to 0.606)	0.533 (0.427 to 0.623)	0.647 (0.575 to 0.717)
Observed event proportion (95% CI)	0.657 (0.517 to 0.767)	0.657 (0.517 to 0.767)	0.657 (0.517 to 0.767)

AUROC, AP, and Brier score are point estimates without marginal CIs in this panel. Calibration and prediction-summary intervals are reported where available.

C. RBF-SVM versus MELD-Na: paired differences

Cohort	AUROC difference (95% CI)	AP difference (95% CI)	Brier improvement (95% CI)
Internal validation	0.060 (-0.021 to 0.141)	0.052 (-0.050 to 0.159)	0.019 (-0.011 to 0.048)
External evaluation	0.100 (-0.029 to 0.233)	0.122 (0.001 to 0.230)	0.015 (-0.036 to 0.072)

AUROC and AP differences are RBF-SVM minus MELD-Na; Brier improvement is MELD-Na minus RBF-SVM. Positive values favor RBF-SVM.

C. Other paired comparisons: internal validation

Comparison (left minus right)	AUROC difference (95% CI)	AP difference (95% CI)	Brier difference (95% CI)
Core4 logistic minus MELD-Na benchmark	0.071 (-0.013 to 0.156)	0.096 (0.001 to 0.195)	-0.025 (-0.054 to 0.005)
RBF-SVM minus Core4 logistic	-0.012 (-0.038 to 0.014)	-0.044 (-0.094 to 0.004)	0.006 (-0.004 to 0.016)

All differences are the left model minus the right model. A negative Brier difference favors the left model.

C. Other paired comparisons: external evaluation

Comparison (left minus right)	AUROC difference (95% CI)	AP difference (95% CI)	Brier difference (95% CI)
Core4 logistic minus MELD-Na benchmark	0.042 (-0.082 to 0.158)	0.054 (-0.043 to 0.140)	0.011 (-0.041 to 0.065)
RBF-SVM minus Core4 logistic	0.058 (-0.010 to 0.143)	0.068 (-0.018 to 0.169)	-0.026 (-0.060 to -0.000)

All differences are the left model minus the right model. A negative Brier difference favors the left model.

D. Nested patient-level aggregate: coverage

Patients	Events	Paired partitions	Total paired predictions
445	185	98	13034

D. Paired held-out predictions per patient

Minimum	Q1	Median	Mean	Q3	Maximum
15	26	29	29.2899	32	45

D. Nested patient-level aggregate: absolute and paired performance

Metric	RBF-SVM (95% CI)	MELD-Na benchmark (95% CI)	Paired difference (95% CI)
AUROC	0.783 (0.739 to 0.824)	0.722 (0.675 to 0.769)	0.061 (0.025 to 0.097)
AP	0.679 (0.606 to 0.760)	0.618 (0.546 to 0.691)	0.061 (0.012 to 0.116)
Brier	0.189 (0.172 to 0.206)	0.212 (0.196 to 0.228)	0.024 (0.010 to 0.036)

AUROC and AP differences are RBF-SVM minus MELD-Na; Brier improvement is MELD-Na minus RBF-SVM. Positive values favor RBF-SVM.

Each patient contributes the mean of paired predictions obtained only while held out. CIs use 5,000 paired patient-bootstrap samples and condition on the aggregated predictions without repeating model development. This aggregate summarizes the Chinese cohort rather than an independent cohort or a newly fitted single model.

Fixed internal validation comprised 133 patients with 55 events. External evaluation comprised 102 episodes from 50 patients, including 67 event episodes. AUROC, AP, and Brier score in panel B are point estimates; calibration and prediction summaries include their available intervals.

Fixed paired comparisons resample patients internally and patient clusters externally. Diagnostic calibration does not replace the evaluated predictions. Further details are in Notes M2 and M4.

Abbreviations: AUROC, area under the receiver operating characteristic curve; AP, average precision; CITL, calibration-in-the-large; CI, confidence interval; O/E, observed-to-expected ratio; Q1/Q3, first/third quartiles; LASSO, least absolute shrinkage and selection operator; RBF-SVM, radial-basis-function support vector machine; XGBoost, Extreme Gradient Boosting; LightGBM, Light Gradient Boosting Machine; MELD-Na, Model for End-Stage Liver Disease-sodium. Core4 comprises international normalized ratio, platelet count, albumin-bilirubin score and fibrosis-4 index.

E. External model-ranking stability under patient-cluster bootstrap resampling

AUROC

Model	Mean rank	First-place share (%)	Top-three share (%)
Logistic regression	4.38	1.13	28.19
LASSO	5.41	0.11	13.72
Elastic net	3.59	6.40	48.75
Random forest	4.11	0.31	49.78
XGBoost	3.31	4.43	62.11
RBF-SVM	1.24	87.62	95.61
LightGBM	5.96	0.00	1.84

AP

Model	Mean rank	First-place share (%)	Top-three share (%)
Logistic regression	4.86	1.00	14.76
LASSO	6.16	0.02	5.40
Elastic net	4.34	6.36	25.00
Random forest	3.36	1.78	73.60
XGBoost	2.24	28.74	86.92
RBF-SVM	1.73	62.10	91.80
LightGBM	5.31	0.00	2.52

Brier score

Model	Mean rank	First-place share (%)	Top-three share (%)
Logistic regression	5.16	0.80	12.36
LASSO	5.42	0.14	14.06
Elastic net	4.11	4.60	32.60
Random forest	3.55	0.92	61.00
XGBoost	3.35	2.80	65.62
RBF-SVM	1.20	89.54	97.36
LightGBM	5.21	1.20	17.00

Rank summaries use 5,000 common patient-cluster bootstrap samples of fixed external predictions for 102 episodes from 50 patients (67 event episodes). Each sample draws 50 patients with replacement and retains all their eligible episodes with the corresponding multiplicities; estimates remain episode-weighted. Models are ranked separately by higher AUROC, higher AP, and lower Brier score, with 1 indicating the best rank. Tied ranks are equally apportioned using a tolerance of 10^{-12}. Mean rank is averaged across samples; first-place and top-three shares sum the corresponding apportioned frequencies. All 5,000 samples were valid for each metric. Shares describe resampling rankings and are not probabilities that a model is truly optimal.

Abbreviations: AUROC, area under the receiver operating characteristic curve; AP, average precision; LASSO, least absolute shrinkage and selection operator; RBF-SVM, radial-basis-function support vector machine; XGBoost, Extreme Gradient Boosting; LightGBM, Light Gradient Boosting Machine.

F. Paired external performance differences between RBF-SVM and the other candidate models

ΔAUROC (RBF-SVM minus comparator)

Comparator	Point difference (95% bootstrap CI)	RBF-SVM-leading frequency (%)
Logistic regression	+0.0580 (-0.0104 to 0.1435)	94.68
LASSO	+0.0648 (-0.0068 to 0.1551)	95.94
Elastic net	+0.0525 (-0.0167 to 0.1381)	91.94
Random forest	+0.0503 (0.0042 to 0.1173)	98.66
XGBoost	+0.0420 (-0.0083 to 0.1112)	94.32
LightGBM	+0.0753 (0.0221 to 0.1458)	99.80

ΔAP (RBF-SVM minus comparator)

Comparator	Point difference (95% bootstrap CI)	RBF-SVM-leading frequency (%)
Logistic regression	+0.0678 (-0.0182 to 0.1691)	91.94
LASSO	+0.0774 (-0.0149 to 0.1793)	93.80
Elastic net	+0.0656 (-0.0231 to 0.1680)	90.08
Random forest	+0.0193 (-0.0176 to 0.0948)	85.08
XGBoost	+0.0044 (-0.0326 to 0.0884)	67.32
LightGBM	+0.0544 (0.0056 to 0.1429)	99.00

Brier improvement (comparator minus RBF-SVM)

Comparator	Point difference (95% bootstrap CI)	RBF-SVM-leading frequency (%)
Logistic regression	+0.0256 (0.0000243 to 0.0600)	97.52
LASSO	+0.0262 (-0.0015 to 0.0608)	96.66
Elastic net	+0.0224 (-0.0057 to 0.0555)	94.20
Random forest	+0.0161 (0.0003 to 0.0336)	97.64
XGBoost	+0.0156 (-0.0021 to 0.0349)	95.94
LightGBM	+0.0264 (0.0019 to 0.0483)	98.16

Point differences use the complete external cohort. Paired 95% percentile CIs are the 2.5th and 97.5th percentiles of within-sample differences from the same 5,000 patient-cluster samples as panel E. Differences are RBF-SVM minus comparator for AUROC and AP and comparator minus RBF-SVM for Brier score; positive values favor RBF-SVM. The RBF-SVM-leading frequency is the proportion of valid samples with a difference $>10^{-12}$, retaining ties in the denominator; it is not a P value. All samples were valid. CIs are comparison-specific and unadjusted for multiplicity; additional digits are retained for near-zero bounds. Models were not refitted. These exploratory summaries do not provide independent validation or adjust for primary-model selection.

Abbreviations: CI, confidence interval; AUROC, area under the receiver operating characteristic curve; AP, average precision; LASSO, least absolute shrinkage and selection operator; RBF-SVM, radial-basis-function support vector machine; XGBoost, Extreme Gradient Boosting; LightGBM, Light Gradient Boosting Machine.

Table S6 Predictor representation and Core4 ablation analyses

A. Internal validation: Logistic regression

Representation	AUROC	AP	Brier	CITL	Slope
Core4 reference set	0.762	0.716	0.193	-0.063	0.796
Raw-only intersection	0.755	0.715	0.197	-0.118	0.762
Composite representation	0.744	0.676	0.202	0.064	0.926

A. Internal validation: RBF-SVM

Representation	AUROC	AP	Brier	CITL	Slope
Core4 reference set	0.751	0.672	0.199	0.020	0.822
Raw-only intersection	0.763	0.724	0.191	-0.064	0.873
Composite representation	0.739	0.667	0.204	0.044	0.970

A. External evaluation: Logistic regression

Representation	AUROC	AP	Brier	CITL	Slope
Core4 reference set	0.712	0.785	0.225	0.781	0.223
Raw-only intersection	0.665	0.744	0.249	0.825	0.124
Composite representation	0.719	0.797	0.214	0.601	0.420

A. External evaluation: RBF-SVM

Representation	AUROC	AP	Brier	CITL	Slope
Core4 reference set	0.770	0.853	0.199	0.725	0.853
Raw-only intersection	0.675	0.765	0.243	0.684	0.240
Composite representation	0.733	0.834	0.209	0.582	0.755

B. Core4 ablations: internal validation / Logistic regression

Ablation	Delta AUROC (95% CI)	Delta AP (95% CI)	Brier improvement (95% CI)
E1: remove INR	0.030 (-0.007 to 0.067)	0.063 (0.004 to 0.117)	0.015 (0.001 to 0.029)
E2: remove PLT	-0.006 (-0.043 to 0.029)	-0.002 (-0.037 to 0.034)	0.001 (-0.012 to 0.013)
E3: remove ALBI	0.011 (-0.015 to 0.037)	-0.001 (-0.035 to 0.034)	0.005 (-0.004 to 0.014)
E4: remove FIB-4	-0.011 (-0.025 to 0.002)	-0.016 (-0.046 to 0.009)	-0.002 (-0.008 to 0.003)
E5: INR + ALBI only	-0.024 (-0.078 to 0.025)	-0.029 (-0.080 to 0.019)	-0.008 (-0.027 to 0.010)
E6: PLT + FIB-4 only	0.095 (0.037 to 0.158)	0.149 (0.058 to 0.224)	0.034 (0.013 to 0.054)

B. Core4 ablations: internal validation / RBF-SVM

Ablation	Delta AUROC (95% CI)	Delta AP (95% CI)	Brier improvement (95% CI)
E1: remove INR	0.037 (-0.004 to 0.078)	0.056 (-0.011 to 0.118)	0.016 (-0.002 to 0.033)
E2: remove PLT	-0.003 (-0.025 to 0.017)	-0.023 (-0.075 to 0.028)	-0.002 (-0.010 to 0.007)
E3: remove ALBI	0.010 (-0.022 to 0.043)	-0.026 (-0.079 to 0.028)	0.004 (-0.008 to 0.017)
E4: remove FIB-4	-0.024 (-0.053 to 0.005)	-0.043 (-0.110 to 0.020)	-0.010 (-0.024 to 0.003)
E5: INR + ALBI only	-0.036 (-0.082 to 0.008)	-0.072 (-0.139 to -0.005)	-0.012 (-0.031 to 0.006)
E6: PLT + FIB-4 only	0.090 (0.026 to 0.155)	0.114 (0.022 to 0.196)	0.032 (0.008 to 0.056)

B. Core4 ablations: external evaluation / Logistic regression

Ablation	Delta AUROC (95% CI)	Delta AP (95% CI)	Brier improvement (95% CI)
E1: remove INR	-0.049 (-0.133 to 0.030)	-0.084 (-0.172 to 0.008)	0.002 (-0.041 to 0.040)
E2: remove PLT	0.005 (-0.042 to 0.055)	0.013 (-0.026 to 0.058)	-0.007 (-0.028 to 0.014)
E3: remove ALBI	0.032 (0.006 to 0.065)	0.033 (0.004 to 0.070)	0.012 (-0.001 to 0.028)
E4: remove FIB-4	0.008 (-0.010 to 0.029)	0.010 (-0.011 to 0.035)	0.009 (-0.001 to 0.017)
E5: INR + ALBI only	0.056 (-0.021 to 0.138)	0.058 (-0.012 to 0.126)	0.018 (-0.018 to 0.053)
E6: PLT + FIB-4 only	-0.001 (-0.105 to 0.100)	-0.053 (-0.145 to 0.055)	0.013 (-0.034 to 0.055)

B. Core4 ablations: external evaluation / RBF-SVM

Ablation	Delta AUROC (95% CI)	Delta AP (95% CI)	Brier improvement (95% CI)
E1: remove INR	-0.004 (-0.059 to 0.055)	-0.022 (-0.054 to 0.032)	-0.003 (-0.028 to 0.022)
E2: remove PLT	0.025 (-0.035 to 0.099)	0.016 (-0.045 to 0.123)	0.003 (-0.019 to 0.031)
E3: remove ALBI	0.072 (-0.007 to 0.166)	0.083 (-0.018 to 0.189)	0.016 (-0.017 to 0.061)
E4: remove FIB-4	0.016 (-0.028 to 0.061)	0.006 (-0.034 to 0.055)	0.010 (-0.009 to 0.027)
E5: INR + ALBI only	0.107 (0.009 to 0.221)	0.105 (-0.010 to 0.218)	0.038 (-0.001 to 0.079)
E6: PLT + FIB-4 only	0.034 (-0.039 to 0.125)	-0.003 (-0.041 to 0.082)	0.009 (-0.022 to 0.044)

Panel A reports point estimates for fixed internal validation (133 patients) and external evaluation (102 episodes from 50 patients). Selected intersections were INR/PLT/ALBI/FIB-4 for Core4, INR/PLT for raw-variable candidates, and PTAR/FIB-4 for composite candidates. Candidate-pool definitions are in Note M3.

Panel B reports all six ablations. ΔAUROC and ΔAP are full model minus ablation; Brier improvement is ablation minus full model. Positive values favor the full model. Paired 95% percentile CIs use 5,000 patient-bootstrap samples internally and patient-cluster samples externally. Intervals crossing zero do not establish equivalence or absence of contribution. Details are in Note M3.

Abbreviations: AUROC, area under the receiver operating characteristic curve; AP, average precision; CITL, calibration-in-the-large; CI, confidence interval; RBF-SVM, radial-basis-function support vector machine; PTAR, study-defined 10 × INR/albumin (albumin in g/L); INR, international normalized ratio; PLT, platelet count; ALBI, albumin-bilirubin score; FIB-4, fibrosis-4 index.

Table S7 Discrimination by readmission timing

A. Event-time groups and shared non-event controls

Cohort / period	Cases	Common controls	Analyzed records	Patients / units	Unclassified positive times
Internal / Early	22	78	100	100	0
Internal / Late	33	78	111	111	0
External / Early	48	35	83	48	2
External / Late	17	35	52	37	2

Internal units are patients, and external units are hospitalization episodes. The same 1-year non-event controls are shared by the early and later comparisons; overlapping comparison sizes cannot be added to reconstruct the primary cohort.

Two external event episodes with unclassifiable timing remain in the annual analysis but are excluded from both timing groups. The repeated count in the external rows refers to the same two episodes.

B. Fixed internal model scores: time-stratified discrimination

Model	Early AUROC (95% CI)	Late AUROC (95% CI)	Late minus early (95% CI)
Logistic regression	0.846 (0.745 to 0.930)	0.707 (0.591 to 0.814)	-0.139 (-0.270 to -0.012)
LASSO	0.848 (0.749 to 0.932)	0.711 (0.596 to 0.817)	-0.137 (-0.267 to -0.011)
Elastic net	0.843 (0.744 to 0.929)	0.709 (0.594 to 0.814)	-0.134 (-0.261 to -0.009)
Random forest	0.837 (0.736 to 0.924)	0.711 (0.601 to 0.815)	-0.126 (-0.253 to -0.004)
XGBoost	0.838 (0.736 to 0.926)	0.686 (0.572 to 0.793)	-0.152 (-0.285 to -0.026)
RBF-SVM	0.818 (0.712 to 0.911)	0.706 (0.593 to 0.811)	-0.112 (-0.235 to 0.012)

C. Fixed external model scores: time-stratified discrimination

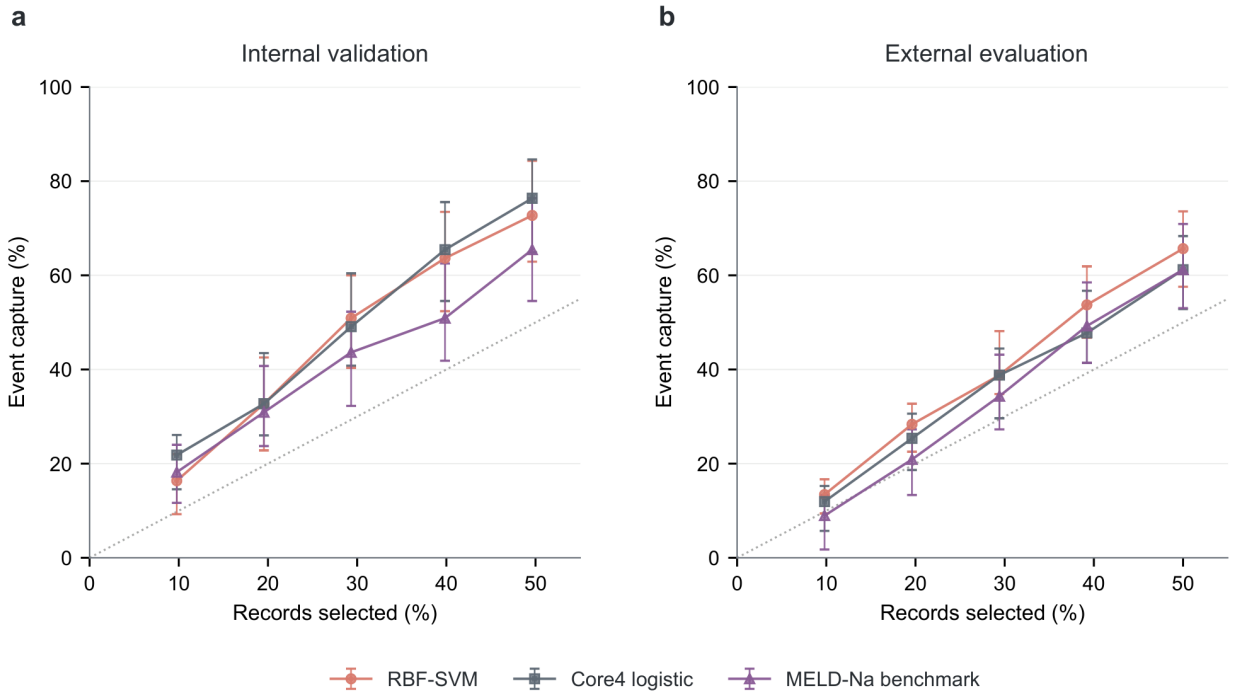
Model	Early AUROC (95% CI)	Late AUROC (95% CI)	Late minus early (95% CI)
Logistic regression	0.725 (0.583 to 0.835)	0.645 (0.438 to 0.803)	-0.080 (-0.243 to 0.048)
LASSO	0.718 (0.577 to 0.829)	0.640 (0.436 to 0.798)	-0.078 (-0.241 to 0.051)
Elastic net	0.732 (0.592 to 0.842)	0.649 (0.444 to 0.807)	-0.083 (-0.255 to 0.047)
Random forest	0.728 (0.574 to 0.847)	0.684 (0.447 to 0.853)	-0.044 (-0.232 to 0.112)
XGBoost	0.732 (0.569 to 0.853)	0.691 (0.456 to 0.857)	-0.042 (-0.227 to 0.109)
RBF-SVM	0.786 (0.655 to 0.887)	0.723 (0.494 to 0.878)	-0.064 (-0.266 to 0.077)

Early events occurred at 0–90 days, and later events at >90–365 days. Each event group is compared with the same 1-year non-event controls; the other event group is excluded. These analyses evaluate six fixed annual model scores, rather than newly fitted early/later models, cumulative 90-day ROC curves, or time-specific calibrated risks.

CIs and later-minus-earlier contrasts use 5,000 paired resamples of patients internally and patient clusters externally. An interval crossing zero does not establish equivalence. Details are in Note M3.

Abbreviations: AUROC, area under the receiver operating characteristic (ROC) curve; CI, confidence interval; LASSO, least absolute shrinkage and selection operator; RBF-SVM, radial-basis-function support vector machine; XGBoost, Extreme Gradient Boosting.

Fig. S1 Observed-event capture at matched selection capacity



(a) Fixed internal validation (133 patients) and (b) external evaluation (102 hospitalization episodes from 50 patients). RBF-SVM, Core4 logistic regression, and the MELD-Na benchmark are compared at nominal selection proportions of 10%, 20%, 30%, 40%, and 50%. Each selects the K highest-scoring observations, where K = floor(qN), q is nominal capacity, and N is the cohort record count. The horizontal axis shows actual coverage (K/N); the vertical axis shows the proportion of all observed annual event records captured. Points and bars are estimates and pointwise 95% percentile CIs from 5,000 resamples of patients internally and patient clusters externally. Top K selection is repeated within each resample; ties use the fixed outcome-blind identifier hash. Lines join the five scenarios; the dotted diagonal shows expected capture under random ranking. Core4 comprises INR, PLT, ALBI, and FIB-4. MELD-Na is a development-fitted single-predictor logistic benchmark. These hypothetical allocation scenarios differ from the development-derived risk strata. Intervals condition on fixed predictions, and capture does not estimate prevented events or an intervention effect

RBF-SVM, radial-basis-function support vector machine; MELD-Na, Model for End-Stage Liver Disease-sodium; CI, confidence interval; INR, international normalized ratio; PLT, platelet count; ALBI, albumin-bilirubin score; FIB-4, fibrosis-4 index

Fig. S2 Full-range decision curves and paired net-benefit differences

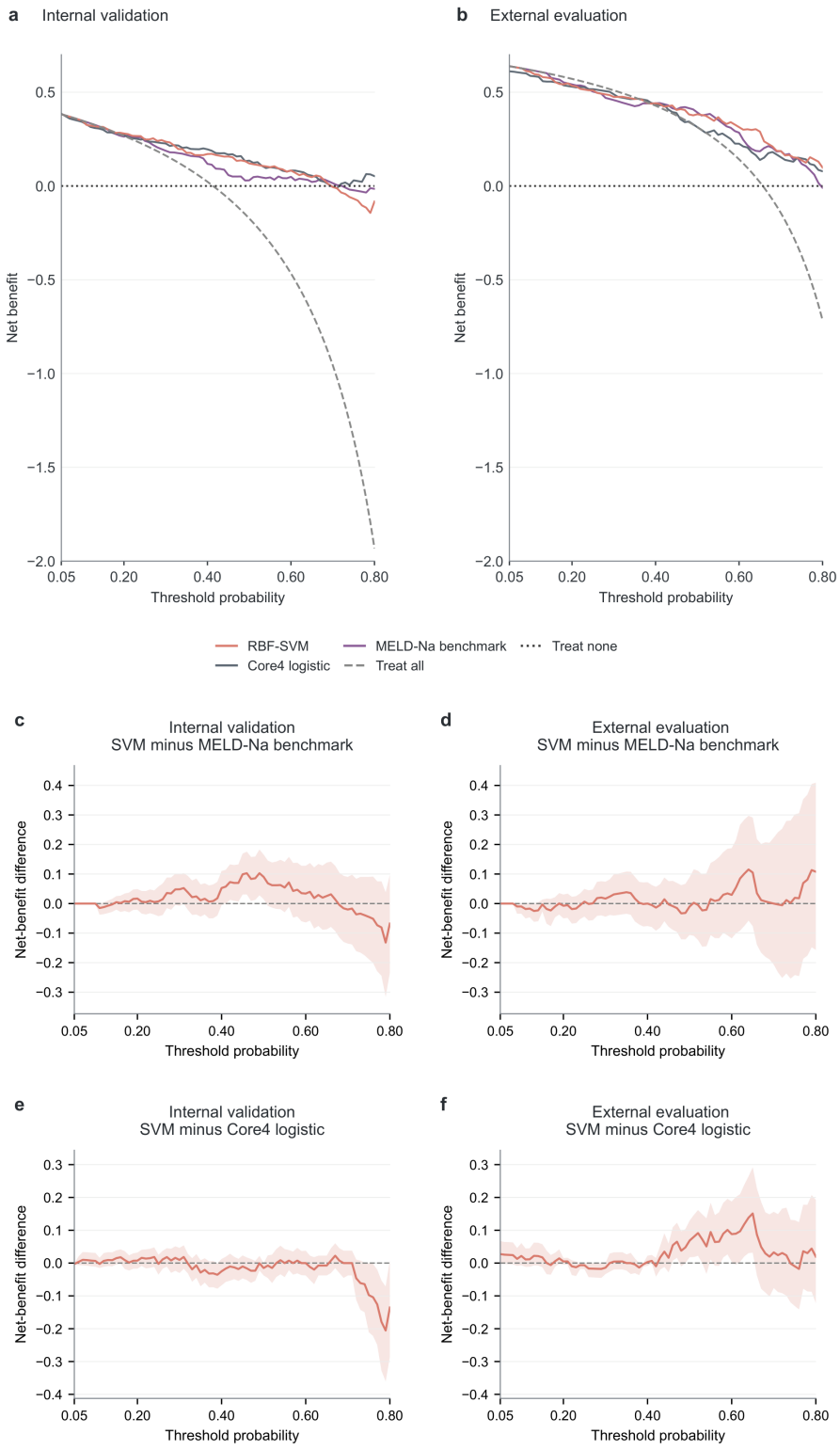


Fig. S2 (a, b) Decision curves in fixed internal validation (133 patients) and external evaluation (102 episodes from 50 patients). Solid curves show net benefit for RBF-SVM, Core4 logistic regression, and the MELD-Na benchmark; dashed gray and dotted zero lines denote treat-all and treat-none. These panels use identical axes and show point estimates. (c, d) RBF-SVM minus MELD-Na net-benefit differences internally and externally. (e, f) Corresponding RBF-SVM minus Core4 logistic-regression differences. All panels use thresholds from 0.05 to 0.80 at increments of 0.01. Shading in c–f shows pointwise 95% percentile CIs of paired differences from 5,000 patient resamples internally and patient-cluster resamples externally. Dashed

horizontal lines denote zero; positive differences favor RBF-SVM at that threshold. Intervals are pointwise rather than simultaneous across thresholds. Core4 comprises INR, PLT, ALBI, and FIB-4; the MELD-Na benchmark is a development-fitted single-predictor logistic model. Net benefit weights false-positive classifications by threshold odds. Comparisons condition on fixed predictions, without evaluation-cohort recalibration or threshold optimization. Treat-all and treat-none refer to a hypothetical risk-triggered action; the displayed range does not establish validated intervention thresholds or demonstrated clinical benefit

DCA, decision-curve analysis; RBF-SVM, radial-basis-function support vector machine (SVM in the panels); MELD-Na, Model for End-Stage Liver Disease-sodium; CI, confidence interval; INR, international normalized ratio; PLT, platelet count; ALBI, albumin-bilirubin score; FIB-4, fibrosis-4 index

Table S8 Baseline characteristics of the Chinese cohort

Characteristic	Overall cohort (n = 445)
Demographic and background characteristics	
Age (years)	52.0 (45.0, 61.0)
Male sex	313 (70.3)
BMI (kg/m²)	22.49 (20.55, 24.44)
Obesity, n (%)	35 (7.9)
Hypertension, n (%)	51 (11.5)
Diabetes mellitus, n (%)	79 (17.8)
Metabolic comorbidities, n (%)	131 (29.4)
Smoking history	185 (41.6)
Alcohol use history	183 (41.1)
HBV serological, virological and treatment-related characteristics	
HBeAg/anti-HBe status	
HBeAg−/anti-HBe+	313 (70.3)
HBeAg+/anti-HBe−	77 (17.3)
HBeAg−/anti-HBe−	36 (8.1)
HBeAg+/anti-HBe+	19 (4.3)
HBV DNA study category (IU/mL)	Available n = 442
<2,000	260 (58.8)
2,000–<20,000	47 (10.6)
≥20,000	135 (30.5)
HBV DNA, log10(IU/mL)	2.70 (2.70, 4.97); n = 437
HBsAg, log10(IU/mL)	2.81 (1.94, 3.16)
Prior antiviral therapy	246 (55.3)
Recorded clinical manifestations	
Ascites	194 (43.6)
EGVH	246 (55.3)
SBP	108 (24.3)
HE	27 (6.1)
HRS	5 (1.1)

Characteristic	Overall cohort (n = 445)
Infection	181 (40.7)
Laboratory measurements	
Albumin (g/L)	29.00 (26.00, 34.00)
Total bilirubin (μmol/L)	24.70 (15.70, 41.00)
ALT (U/L)	39.00 (27.00, 57.00)
AST (U/L)	44.00 (30.00, 65.00)
ALP (U/L)	92.00 (67.00, 128.00)
GGT (U/L)	38.00 (24.00, 74.00)
INR	1.37 (1.22, 1.52)
PT (s)	16.40 (15.10, 18.30)
PTA (%)	61.00 (52.00, 71.00)
Platelet count (109/L)	63.00 (44.00, 95.00)
WBC (10^9/L)	3.97 (2.70, 5.95)
Hemoglobin (g/L)	92.00 (73.00, 111.00)
Serum creatinine (μmol/L)	67.00 (56.00, 79.00)
Sodium (mmol/L)	139.00 (137.00, 141.00)
Composite scores	
MELD-Na	12.47 (9.99, 15.54)
ALBI	-1.62 (-2.02, -1.22)
FIB-4	5.76 (3.38, 10.53)
Index hospitalization	
Length of index hospital stay (days)	10.0 (7.0, 15.0)
Observed 1-year outcome	
Target readmission	185 (41.6)

Values are medians (interquartile ranges) or n (%), calculated from available nonmissing observations. Available n is shown when it differs from 445. Clinical manifestations can coexist. Metabolic comorbidity denotes obesity, hypertension, or diabetes mellitus; obesity was defined as BMI ≥28 kg/m². Viral-marker coding, measurement limitations, and derived-index definitions are detailed in Table S3A and Note M1.

Abbreviations: ALBI, albumin-bilirubin score; ALP, alkaline phosphatase; ALT, alanine aminotransferase; anti-HBe, antibody to hepatitis B e antigen; AST, aspartate aminotransferase; BMI, body mass index; DNA, deoxyribonucleic acid; EGVH, esophagogastric variceal hemorrhage; FIB-4, fibrosis-4 index; GGT, gamma-glutamyl transferase; HBeAg, hepatitis B e antigen; HBsAg, hepatitis B surface antigen; HBV, hepatitis B virus; HE, hepatic encephalopathy; HRS, hepatorenal syndrome; INR, international normalized ratio; MELD-Na, Model for End-Stage Liver Disease incorporating serum sodium; PT, prothrombin time; PTA, prothrombin activity; SBP, spontaneous bacterial peritonitis; WBC, white blood cell count.

Table S9 Coverage and observed-event capture across RBF-SVM risk strata

Cohort	Risk-stratum strategy	Selected records, n/N (%)	Captured events, n/total events (%)	Missed events, n	Specificity, % (95% CI)
Internal validation	High only	39/133 (29.3%)	28/55 (50.9%)	27	85.9 (77.6–93.0)
Internal validation	Intermediate plus High	89/133 (66.9%)	48/55 (87.3%)	7	47.4 (36.2–58.4)
External evaluation	High only	52/102 (51.0%)	45/67 (67.2%)	22	80.0 (65.8–92.5)
External evaluation	Intermediate plus High	81/102 (79.4%)	59/67 (88.1%)	8	37.1 (20.6–54.8)

Risk strata use the lower and upper tertiles of development-cohort RBF-SVM predictions, applied unchanged to both evaluation cohorts; exact cutoffs and boundary conventions are in Note M4. Coverage is the proportion of observations selected; event capture is the proportion of all observed target events among selected observations. Specificity is the proportion of non-event observations outside the selected group. External results refer to 102 episodes from 50 patients. Captured events do not estimate prevented readmissions, and these quantile cutoffs are not validated intervention thresholds.

Abbreviations: CI, confidence interval; RBF-SVM, radial-basis-function support vector machine.

Note M6 Extended methods

Study design and population

This single-center retrospective cohort study included patients with HBV-related decompensated cirrhosis hospitalized at the First Affiliated Hospital of Chongqing Medical University between January 2012 and December 2023. Discharge was the prediction time point, and the prediction horizon was the subsequent year.

HBV-related cirrhosis was identified according to the Guidelines for the prevention and treatment of chronic hepatitis B (version 2022) [9], using HBV serology or a documented history of chronic HBV infection together with clinical or imaging evidence of cirrhosis. Index hospitalization records were manually reviewed for eligibility, which required qualifying decompensation. Patients with other major causes of liver disease were excluded. Other exclusion criteria included planned index hospitalization, alcohol-related cirrhosis, autoimmune liver disease, tumors, liver transplantation, incomplete follow-up, and extensive missing clinical data (Fig. 1). Acute-on-chronic liver failure (ACLF) was assessed retrospectively against the Guidelines for the diagnosis and treatment of acute-on-chronic liver failure (2025 version) [10]; patients meeting these criteria were excluded.

The final Chinese cohort comprised 445 patients and was divided in an outcome-stratified 70:30 ratio into development (n = 312) and fixed internal validation (n = 133) cohorts. The study was conducted in accordance with the Declaration of Helsinki. The Medical Research Ethics Review Committee of the First Affiliated Hospital of Chongqing Medical University approved the study and the application for a waiver of informed consent (approval no. 2025-925-01; November 21, 2025).

Outcome and follow-up

The primary outcome was unplanned readmission within 1 year after discharge for variceal bleeding, ascites, hepatic encephalopathy, hepatorenal syndrome (HRS), or spontaneous bacterial peritonitis. Readmissions for planned treatment, investigations, or other elective arrangements were not outcome events.

Readmissions were ascertained through inpatient records, outpatient records, and telephone follow-up through 1 year after discharge, with manual review of the records. Patients with incomplete follow-up and those who died during this period before a target readmission were excluded from the final binary analysis.

Predictor measurement and missing data

We extracted 46 predefined candidate predictors from electronic medical records, covering demographic and lifestyle characteristics, HBV serological and virological measures, cirrhosis-related clinical characteristics, routine laboratory tests, and derived indices (Table S3). Laboratory values used for modeling and clinical benchmarking were the available measurements closest to admission, primarily from admission or the early inpatient period. Variable definitions, measurement scales, analytical roles, and pre-imputation availability are detailed in Table S3 and Note M1.

After the development–validation split, the small amount of missing data in the Chinese cohort was handled by multiple imputation by chained equations. Imputation parameters were estimated from the development cohort alone; internal validation data did not contribute to parameter estimation. The outcome was neither imputed nor used as an imputation predictor. Five completed datasets were generated, and one dataset fixed in advance was used for subsequent model development and evaluation. Derived indices were recalculated after processing their component variables, using the formulas and rounding rules in Table S3A. Estimates were not pooled across completed datasets.

External evaluation cohort

The external cohort was drawn from the Medical Information Mart for Intensive Care IV (MIMIC-IV), version 3.1 [11,12]. Two experienced infectious disease clinicians independently reviewed hospitalization eligibility and target-readmission classification for the external cohort. They applied the clinical inclusion

and exclusion criteria and guideline-based outcome definitions used for the Chinese cohort, integrating available diagnoses, investigation reports, and medical orders recorded in MIMIC-IV. Disagreements were resolved through discussion and consensus. For the 1-year postdischarge assessment, subsequent encounter records and available mortality information in MIMIC-IV were reviewed. Readmission ascertainment was limited to events documented in the database. The analysis unit was an eligible hospitalization episode. A patient could contribute multiple episodes, each with a separately defined 1-year postdischarge outcome window. Model inputs were observed measurements from the same continuous emergency–inpatient encounter, selected as close to admission as available data allowed; no statistical imputation was performed. Note M1 describes the measurement hierarchy and its exceptions. Analyses used episode-level predictions, with patient clustering accounted for in uncertainty estimation.

Feature selection

Categorical variables were encoded using predefined clinical categories, and continuous variables retained their numerical form, with the viral-marker transformations described in Table S3A. Feature selection used only development data. Least absolute shrinkage and selection operator (LASSO) regression and Boruta were applied independently to all 46 candidates; their intersection defined the primary predictor set. LASSO used tenfold cross-validation and the one-standard-error penalty (λ_{1se}), while TentativeRoughFix resolved tentative Boruta selections. The resulting four-measure set, termed Core4, comprised the international normalized ratio (INR), platelet count (PLT), albumin-bilirubin (ALBI) score, and fibrosis-4 (FIB-4) index. Spearman correlations and variance inflation factors (VIFs) were examined after selection, rather than used as selection criteria (Table S3C).

Selection stability was assessed by repeating missing-data handling, LASSO, and Boruta within each outer training set during repeated nested evaluation. We recorded selection frequencies and the Jaccard similarity between each selected intersection and the original Core4 set. Exploratory analyses compared alternative predictor representations and six Core4 ablations (Table S6).

Candidate models and evaluation strategy

Seven models were fitted using the common selected predictor set: unpenalized logistic regression, LASSO, elastic net, random forest, XGBoost, LightGBM, and a radial-basis-function support vector machine (RBF-SVM). Hyperparameter optimization was confined to development data and used tenfold cross-validation repeated five times, with 30 model-based optimization evaluations and area under the receiver operating characteristic curve (AUROC) as the tuning criterion. Unpenalized logistic regression did not undergo a parameter search. Final settings, preprocessing, and available search configurations are reported in Table S4.

Performance was assessed using AUROC, average precision (AP), the Brier score, calibration-in-the-large (CITL), and calibration slope. To assess the full modeling procedure under sample perturbation, we conducted 100 outcome-stratified repeated nested partitions of the Chinese cohort, each with 312 outer training and 133 outer test patients. Missing-data handling, feature selection, hyperparameter optimization, and fitting were repeated within the outer training data. Outer test observations did not inform imputation parameters, feature selection, or hyperparameter choice.

All 100 selection procedures completed. The primary seven-model comparison used 98 compatible partitions; LASSO and elastic net had 95 valid evaluations because three additional single-predictor partitions were unsupported by their fitting interface. Performance distributions were summarized by medians and interquartile ranges across valid outer test sets. These summaries describe variation across overlapping partitions. Execution denominators and exclusions are detailed in Table S5A.

Development-fitted models were applied to the fixed internal and external cohorts without refitting, retuning, or recalibration. The primary model was selected by joint consideration of AUROC, AP, Brier score, CITL, calibration slope, and consistency across the internal and external evaluations, without a prespecified weighted score or a single-metric ranking rule. External performance therefore contributed to model selection. The selected model was used for clinical benchmarking, risk stratification, decision-curve analysis, and prediction attribution.

Clinical benchmarking and risk assessment

Comparators included logistic regression using Core4 and a single-predictor logistic benchmark using continuous MELD-Na. The MELD-Na benchmark was fitted in the development cohort and applied unchanged to the fixed evaluation cohorts. For the nested RBF-SVM–MELD-Na comparison, both models were fitted within the corresponding outer training partitions. Predictions from 98 paired outer test partitions were averaged for each patient using only occasions on which that patient was held out. Paired comparisons of AUROC, AP, and Brier score used these patient-level aggregates (Table S5D). This aggregate is an internal summary of repeated held-out predictions, rather than an additional independent validation cohort.

Low-, intermediate-, and high-risk strata were defined by the lower and upper tertiles of development-cohort RBF-SVM predictions and applied unchanged to the fixed internal and external cohorts. The cutoffs were approximately 0.234 and 0.559, calculated using linear interpolation (type 7 quantiles). We evaluated stratum-specific observed event rates, selected-observation coverage, event capture, and missed events. Calibration was assessed using CITL, calibration slope, and diagnostic calibration curves [13]. Decision-curve analysis (DCA) compared model net benefit across threshold probabilities [14]. Exploratory matched-capacity analyses compared observed-event capture at nominal selection proportions of 10%, 20%, 30%, 40%, and 50%. Exact stratum boundaries, calibration displays, and uncertainty methods are detailed in Note M4.

Exploratory sensitivity analyses examined alternative predictor representations, six Core4 ablations, and discrimination by event timing. For the timing analysis, the existing 1-year model scores were evaluated separately for events occurring within 90 days and after 90 days through day 365, each against the same 1-year non-event controls. These analyses did not fit new time-specific models (Tables S6 and S7).

Prediction attribution

SHapley Additive exPlanations (SHAP) attributed the primary model's predictions to its input features [15]. Exact Kernel SHAP (kernelshap version 0.9.1) quantified contributions on the positive-class probability scale, using all 312 locked development records as the common empirical reference background. All feature coalitions were evaluated without sampling approximation. Global importance was summarized by mean absolute SHAP values; dependence plots related feature values to their contributions. Note M5 documents the background matrix and attribution checks.

Local SHAP decompositions were illustrated using two internal-validation cases with nearly identical MELD-Na scores but different RBF-SVM predictions. The cases were selected after inspection of predictions and outcomes. SHAP values describe attribution within the fitted model, and their interpretation accounts for the shared information between PLT and FIB-4.

Statistical analysis

Continuous variables are summarized as medians and interquartile ranges and were compared with Mann–Whitney U tests. Categorical variables are presented as counts and percentages and were compared with chi-squared or Fisher exact tests. Standardized mean differences (SMDs) described case-mix differences between the Chinese and external cohorts.

Confidence intervals (CIs) for fixed-model performance, paired contrasts, and supplementary decision measures used 5,000 bootstrap resamples. Patients were resampled internally, whereas patient clusters were resampled externally, retaining all eligible hospitalization episodes and the multiplicity of each sampled patient. External point estimates were episode-weighted; clustering accounted for within-patient dependence in their uncertainty. Models shared resamples for paired contrasts, and 95% CIs were calculated from the 2.5th and 97.5th percentiles of valid bootstrap estimates. Risk-stratum event-rate intervals used Wilson intervals internally and patient-cluster bootstrap intervals externally. Calibration-group error bars used bootstrap intervals in both cohorts. Intervals based on fixed or aggregated predictions were conditional on those predictions and did not repeat model development.

All tests were two-sided, with $P < 0.05$ as the nominal significance threshold. Exploratory baseline comparisons, ablations, external model contrasts, and threshold-specific decision analyses were not adjusted for multiplicity. Analyses were performed primarily in R version 4.5.1 and Python version 3.12.14.

Note M7 Extended results

Cohort characteristics

Of 2,538 patients initially identified with HBV-related cirrhosis, 2,093 were excluded during eligibility screening and data curation, leaving 445 in the final analysis: 312 in development and 133 in fixed internal validation (Fig. 1). The primary outcome occurred in 185 patients (41.6%), including 130 development and 55 internal-validation patients. Overall characteristics are reported in Table S8, with outcome-stratified comparisons in Table S1. The external cohort comprised 102 eligible hospitalization episodes from 50 patients, including 67 event episodes (65.7%; Table S2).

In both Chinese subsets, patients subsequently readmitted had lower albumin and platelet counts and higher total bilirubin and INR; MELD-Na, ALBI, and FIB-4 also differed between outcome groups. Differences in sex, hepatitis B surface antigen levels, hepatitis B e antigen/antibody categories, ascites, variceal bleeding, spontaneous bacterial peritonitis, and infection were not statistically significant in either subset. HBV DNA levels and categories differed by outcome in development, but this pattern was not reproduced in fixed internal validation. Complete exploratory comparisons are provided in Table S1.

The external cohort differed from the Chinese cohort in age, INR, creatinine, serum sodium, FIB-4, and MELD-Na, whereas albumin, platelet count, and ALBI were more similar. Median FIB-4 and MELD-Na were 9.65 and 18.43 externally, compared with 5.76 and 12.47 in the Chinese cohort. Full characteristics and SMDs are reported in Table S2.

Complementary selection identified four routine measures

LASSO retained five variables at λ1se, and Boruta retained ten after TentativeRoughFix (Fig. 2). Their intersection comprised INR, PLT, ALBI, and FIB-4. The categorical HBV DNA term was selected by LASSO but not Boruta and was therefore absent from Core4.

The largest absolute Spearman correlation among Core4 measures was 0.74, and VIFs ranged from 1.30 to 1.49, indicating limited residual linear redundancy. Across 100 nested selections, the intersection retained INR in 100%, ALBI in 85%, PLT in 67%, and FIB-4 in 46% of partitions. Median Jaccard similarity to the original Core4 set was 0.75 (interquartile range, 0.50–1.00), indicating that selection stability differed across predictors (Table S3B and C).

The tested alternative predictor representations and Core4 ablations did not identify a replacement or reduced set with consistently better performance across both evaluation cohorts. The raw-variable representation had similar internal point estimates but poorer external performance, and the composite representation showed no consistent advantage. Ablation effects varied by cohort and metric. These exploratory comparisons support retaining Core4 for the present model but do not establish that each measure contributes an independent effect or that reduced models are equivalent (Table S6).

Internal performance was similar across candidate algorithms

Candidate models had broadly similar performance in fixed internal validation (Table 1; Fig. 3a and b). LASSO had AUROC, AP, and Brier score point estimates of 0.766, 0.721, and 0.192, respectively; corresponding values were 0.751, 0.672, and 0.199 for RBF-SVM and 0.747, 0.665, and 0.201 for LightGBM. RBF-SVM had the CITL closest to zero (0.020), although it did not lead the discrimination or overall-error comparisons.

Performance distributions overlapped in repeated nested internal evaluation. Median AUROC was approximately 0.78 for logistic regression, LASSO, elastic net, and RBF-SVM and was slightly lower for random forest, XGBoost, and LightGBM. AP and Brier score showed similar patterns. No algorithm consistently led the internal comparisons across metrics (Fig. 3c and d; Table S5A).

External evaluation differentiated the candidate models more clearly by point estimates. RBF-SVM achieved an AUROC of 0.770 (95% CI, 0.640–0.871), AP of 0.853, Brier score of 0.199, and calibration slope of 0.853. Random forest and XGBoost also had high AP estimates (0.834 and 0.849), but neither had better joint AUROC, Brier score, and calibration slope estimates than RBF-SVM. LightGBM had an AUROC of 0.694, AP of 0.799, and Brier score of 0.225. Across 5,000 patient-cluster bootstrap samples, RBF-SVM had the best mean rank for AUROC, AP, and Brier score. These exploratory rankings and contrasts describe the same fixed external predictions and do not establish uniform superiority (Table S5E and F).

RBF-SVM was selected for subsequent analyses because it combined similar internal performance with favorable external discrimination, probability error, and calibration slope point estimates (Fig. 3; Table 1). This selection reflected joint assessment of multiple metrics and cohorts; the external cohort consequently served both evaluation and model-selection roles.

Comparison with the MELD-Na benchmark

In fixed internal validation, point estimates favored RBF-SVM over MELD-Na: the paired AUROC difference was +0.060 (95% CI, −0.021 to 0.141), AP difference was +0.052 (95% CI, −0.050 to 0.159), and Brier score improvement was +0.019 (95% CI, −0.011 to 0.048) (Fig. 4a; Table S5C).

External differences were +0.100 for AUROC (95% CI, −0.029 to 0.233), +0.122 for AP (95% CI, 0.001 to 0.230), and +0.015 for Brier score improvement (95% CI, −0.036 to 0.072). The AP interval excluded zero, while the AUROC and Brier intervals remained compatible with no difference. Overall, the estimated increments favored RBF-SVM but were imprecise.

Patient-level aggregates of paired held-out predictions from the repeated nested evaluation also favored RBF-SVM: AUROC difference, +0.061 (95% CI, 0.025–0.097); AP difference, +0.061 (95% CI, 0.012–0.116); and Brier score improvement, +0.024 (95% CI, 0.010–0.036). These comparisons provide internal support for the modeling procedure and are conditional on the aggregated predictions (Fig. 4a; Table S5D).

Risk ordering persisted despite external underprediction

RBF-SVM predicted higher risks for observations followed by a target readmission in both evaluation cohorts (Fig. 4b). Applying the development-derived thresholds unchanged, observed event rates increased from low to intermediate to high risk: 15.9%, 40.0%, and 71.8% internally and 38.1%, 48.3%, and 86.5% externally. Risk ordering was preserved, but absolute event rates were higher in every external stratum (Fig. 4c).

The high-risk stratum covered 29.3% of internal-validation patients and captured 50.9% of their events; corresponding external values were 51.0% of episodes and 67.2% of events. Extending selection to the intermediate and high strata increased event capture to 87.3% internally and 88.1% externally, while covering 66.9% of patients and 79.4% of episodes, respectively (Table S9). These strategies quantify the trade-off between broader coverage and fewer missed observed events.

Internal RBF-SVM calibration estimates were a CITL of 0.020 and slope of 0.822. Externally, CITL was 0.725 (95% CI, 0.159–1.272) and slope was 0.853 (95% CI, 0.415–1.491), indicating overall underestimation of risk despite retained risk ordering (Fig. 4d and e; Table S5B).

Decision curves showed positive estimated RBF-SVM net benefit over several displayed threshold ranges in both cohorts (Fig. 4f and g). Matched-capacity analyses also showed enrichment of observed events among higher-ranked observations. Coverage, event capture, full-range decision curves, and paired net-benefit differences are reported in Figs. S1 and S2. These exploratory analyses concern hypothetical risk-triggered actions; they do not measure readmissions prevented through model-guided care.

Discrimination by readmission timing

Discrimination was numerically higher for earlier than later readmissions. In internal validation, 22 events occurred within 90 days and 33 after 90 days through 1 year; AUROCs were 0.818 and 0.706, respectively,

with a later-minus-earlier difference of −0.112 (95% CI, −0.235 to 0.012). Externally, the corresponding 48 and 17 event episodes yielded AUROCs of 0.786 and 0.723 and a difference of −0.064 (95% CI, −0.266 to 0.077). Both timing comparisons used the same cohort-specific 1-year non-event controls, and both difference intervals included zero (Table S7).

SHAP identified consistent model attribution across cohorts

The global attribution ranking was the same in both cohorts: INR, FIB-4, ALBI, and PLT. Mean absolute SHAP values were 0.108, 0.072, 0.056, and 0.032 in fixed internal validation and 0.112, 0.088, 0.059, and 0.039 externally (Fig. 5a and b).

Dependence plots showed nonlinear relationships between Core4 values and their contributions to model output. Higher INR, FIB-4, and ALBI generally contributed to higher predicted probabilities, whereas higher platelet counts generally contributed to lower probabilities (Fig. 5c–f).

Two illustrative internal-validation patients had nearly identical MELD-Na scores but different RBF-SVM predictions. The patient with a target readmission had MELD-Na 16.00 and a predicted probability of 0.803; the patient without an event had MELD-Na 15.99 and a probability of 0.221. Their INR values were 1.66 and 1.21, platelet counts were 36 and 92 × 10^9/L, and FIB-4 values were 12.53 and 3.23, respectively. Local decompositions began from the same SHAP baseline and attributed the divergent predictions to different feature contributions (Fig. 5g and h).